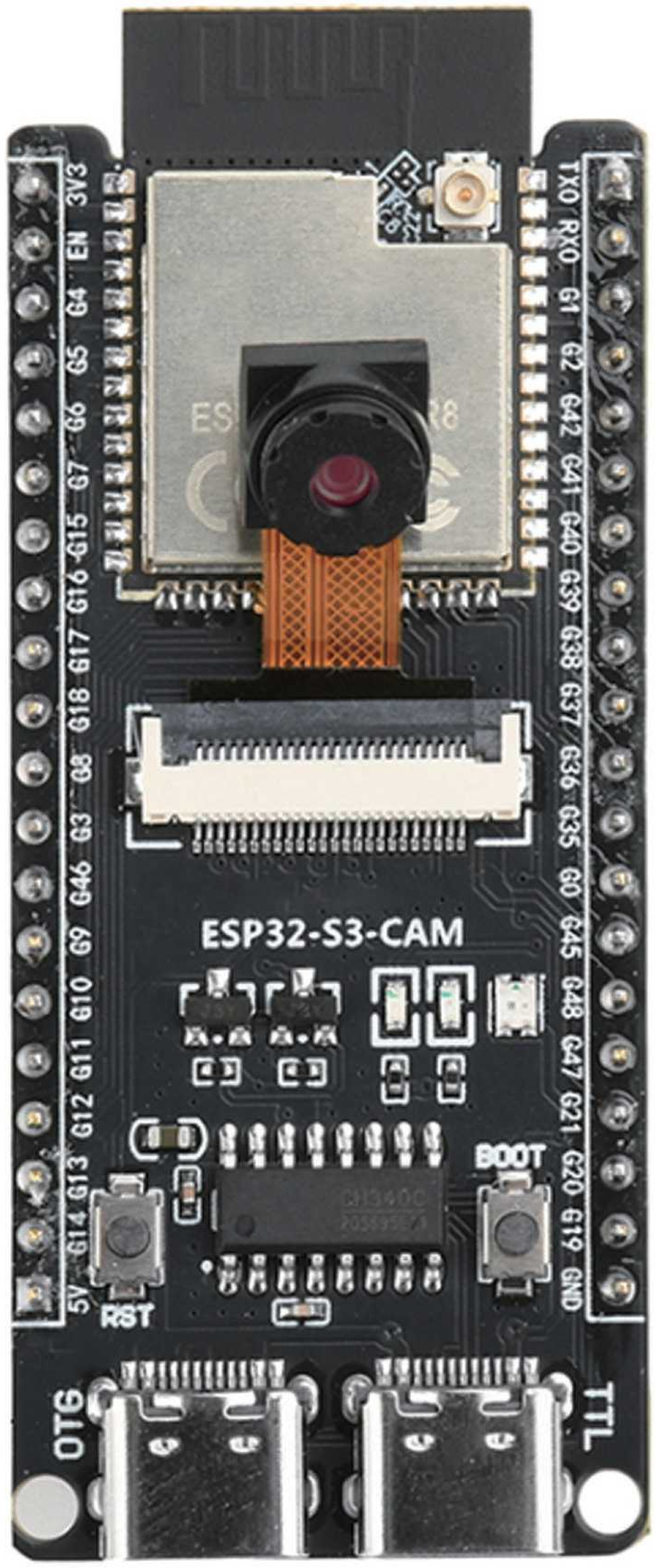
ESP32 S3 CAM Development Board



Catalogue

Product Introduction

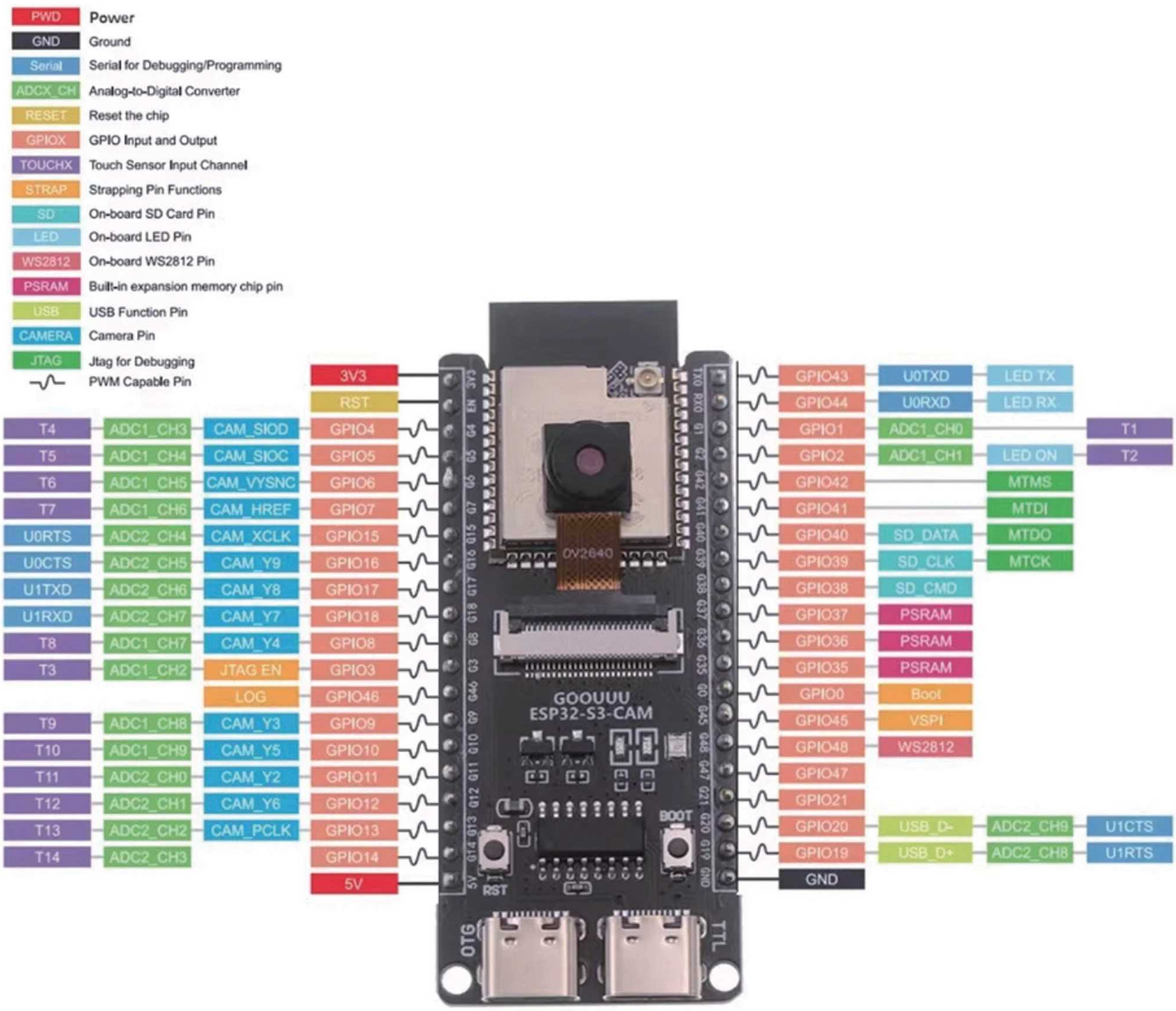
ESP32-S3 integrates 2.4GHz Wi Fi (802.11b/g/n) and supports 40MHz bandwidth; Its low-power Bluetooth subsystem supports Bluetooth 5 (LE) and Bluetooth Mesh, and can achieve long-distance communication through Coded PHY and broadcast extension It also supports 2Mbps PHY to improve transmission speed and data throughput. The ESP32-S3 has superior Wi Fi and Bluetooth LE RF performance and can operate stably even at high temperatures.

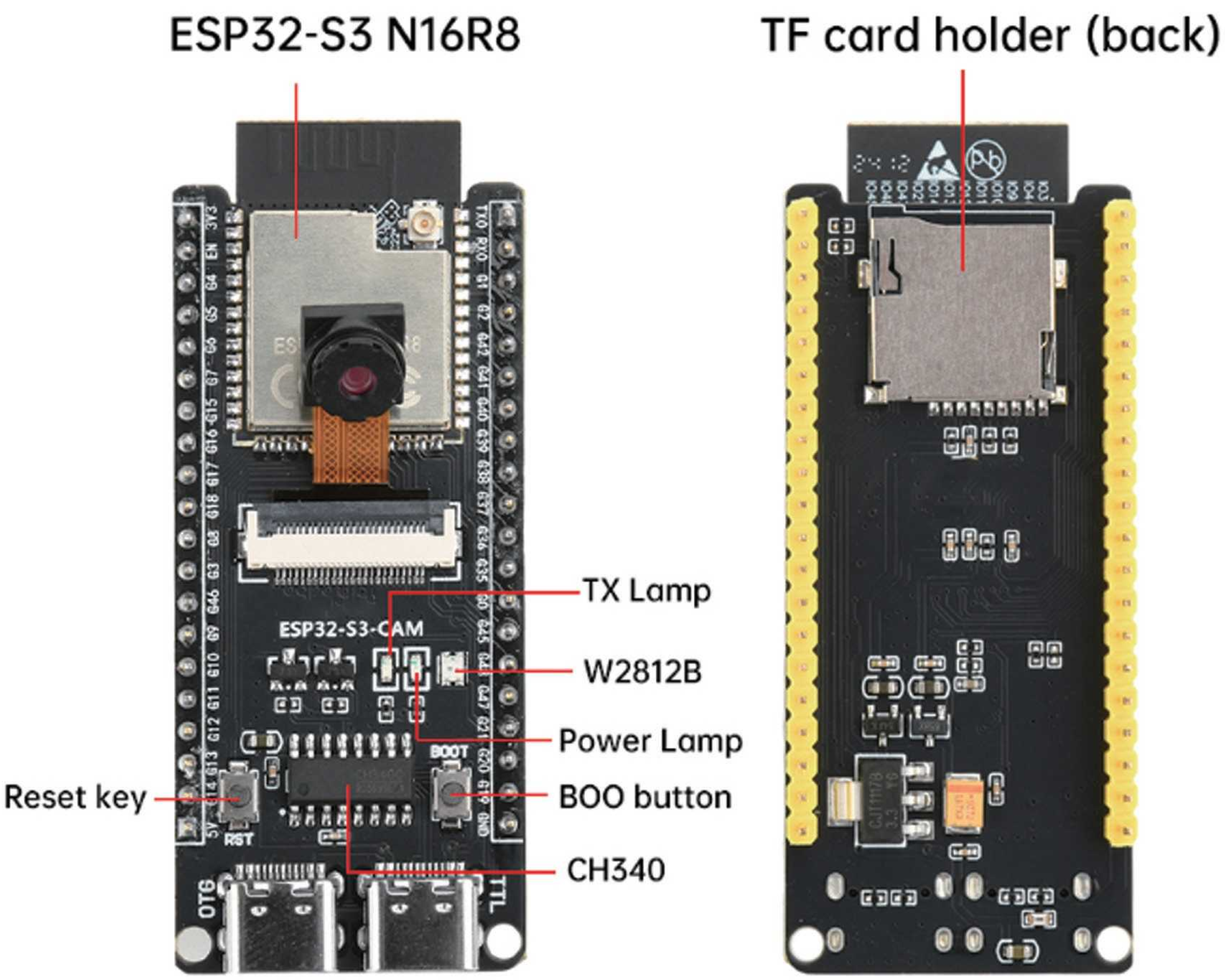
Product features:

1. Excellent Wi Fi performance: Integrated with IEEE 802.11b/g/n protocol to ensure high-speed and stable connection.
2. Advanced Bluetooth subsystem: Supports Bluetooth 5 (LE) and Bluetooth Mesh for long-distance communication.
3. Efficient data processing capability: Built in 32-bit LX7 dual core processor, providing powerful computing performance and a balance of low power consumption.
4. Integration of multiple wireless functions: Wi Fi and Bluetooth LE share the same antenna, optimizing multitasking.
5. High temperature stable operation: With superior performance, it can maintain stable operation even in high temperature environments and is suitable for various harsh working environments.

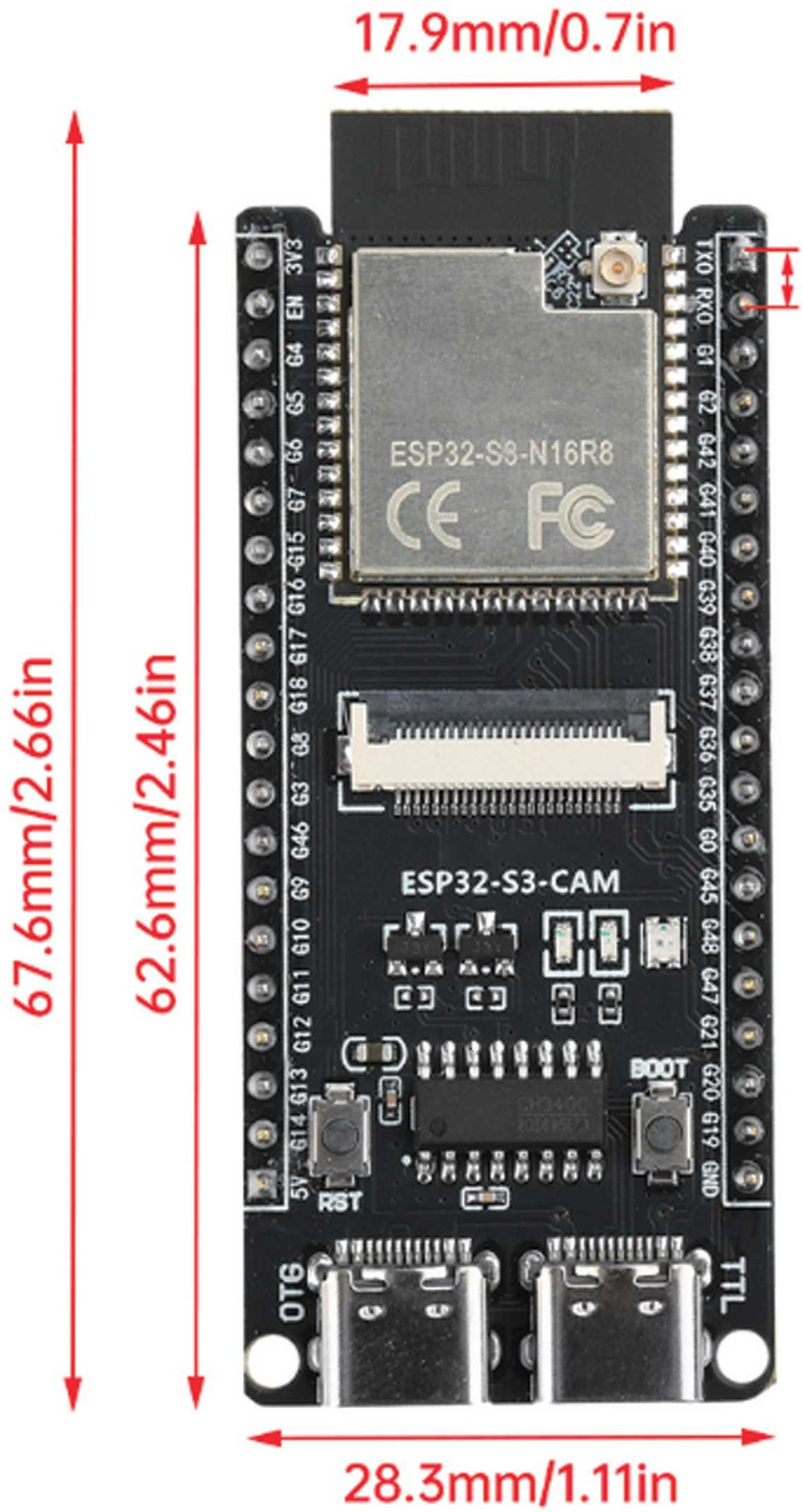
Performance parameters:
Supports IEEE802.11b/g/n protocol
Supports 20MHz and 40MHz bandwidth in the 2.4GHz frequency band
Supports 1T1R mode with a data rate of up to 150Mbps
Wireless Multimedia (WMM)
Frame Aggregation (TX/RX A-MPDU, TX/RX A-MSDU)
Immediate BlockACK
Fragmentation and reassembly
Low power Bluetooth (BluetoothLE): Bluetooth5, BluetoothMesh
High power mode (20dBm)
Supports speeds of 125Kbps, 500Kbps, 1Mbps, 2Mbps
Wi Fi and Bluetooth coexist, sharing the same antenna
32-bit LX7 dual core processor, with a maximum clock speed of 240MHz
WIFI:
Supports IEEE 802.11b/g/n protocol
Supports 20 MHz and 40 MHz bandwidth in the 2.4 GHz band
Supports 1 T1R mode with a data rate of up to 150 Mbps
Wireless Multimedia (WMM)
Frame Aggregation (TX/RX A-MPDU, TX/RX A-MSDU)
Immediate Block ACK
Fragmentation and reassembly
Beacon automatic monitoring (hardware TSF)
4 virtual Wi Fi interfaces
Simultaneously supporting Infrastructure BSS Station mode Please note the SoftAP mode and Station+SoftAP mode, When ESP32-S3 scans in Station mode, The SoftAP channel will change simultaneously
PCB board size: 62.6mm * 28.3mm

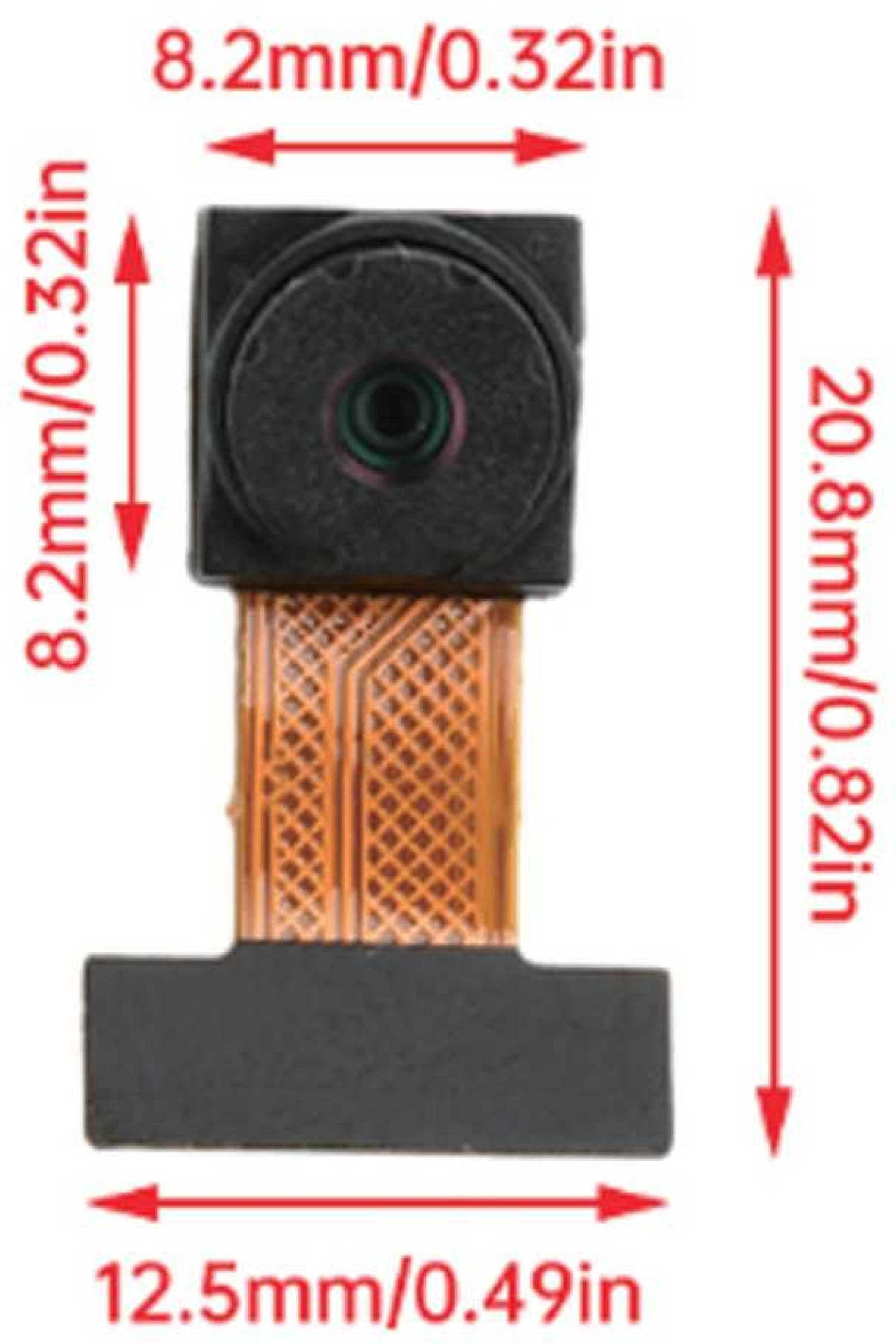
Product Pin Description





Product size



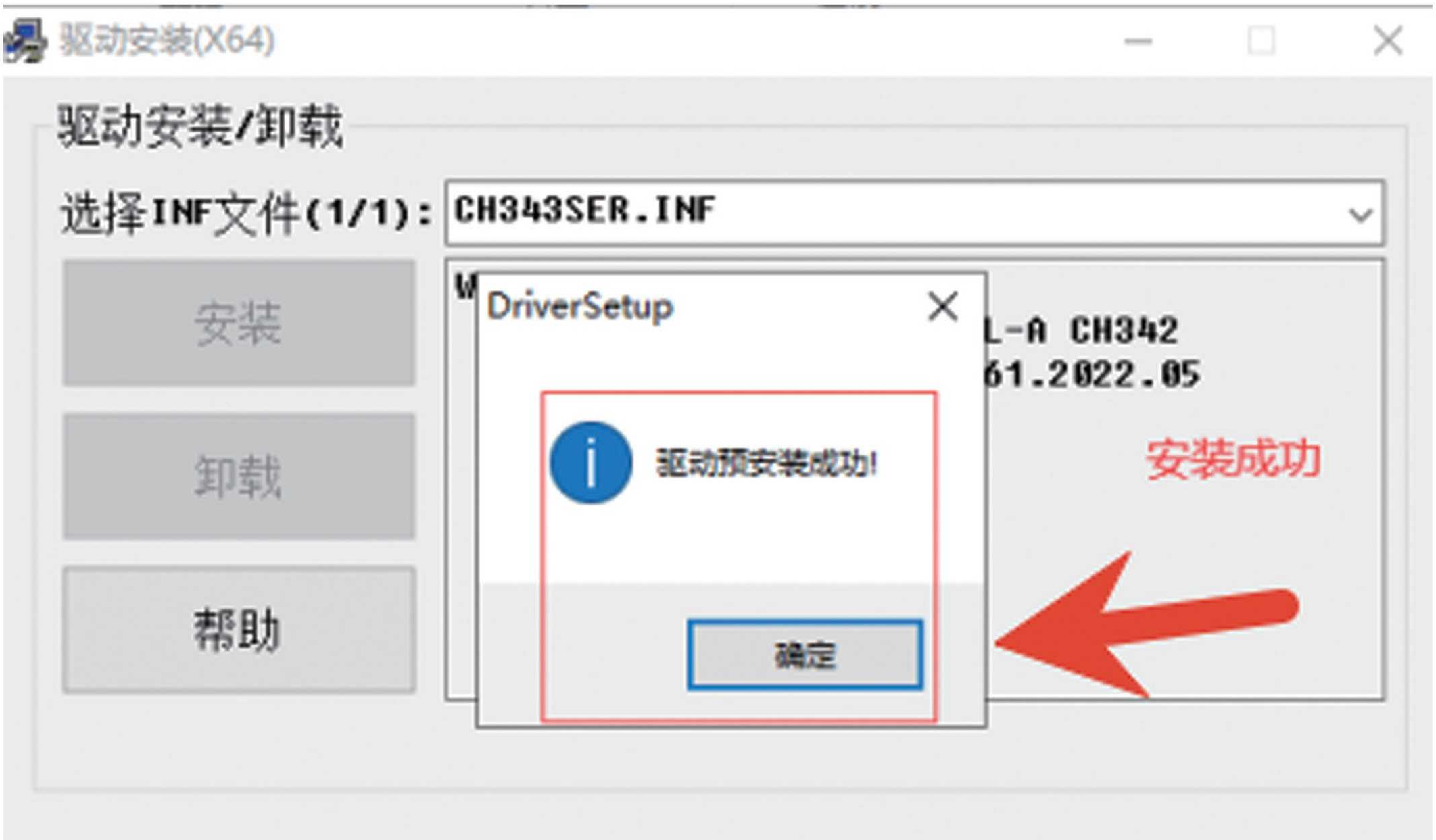


instructions

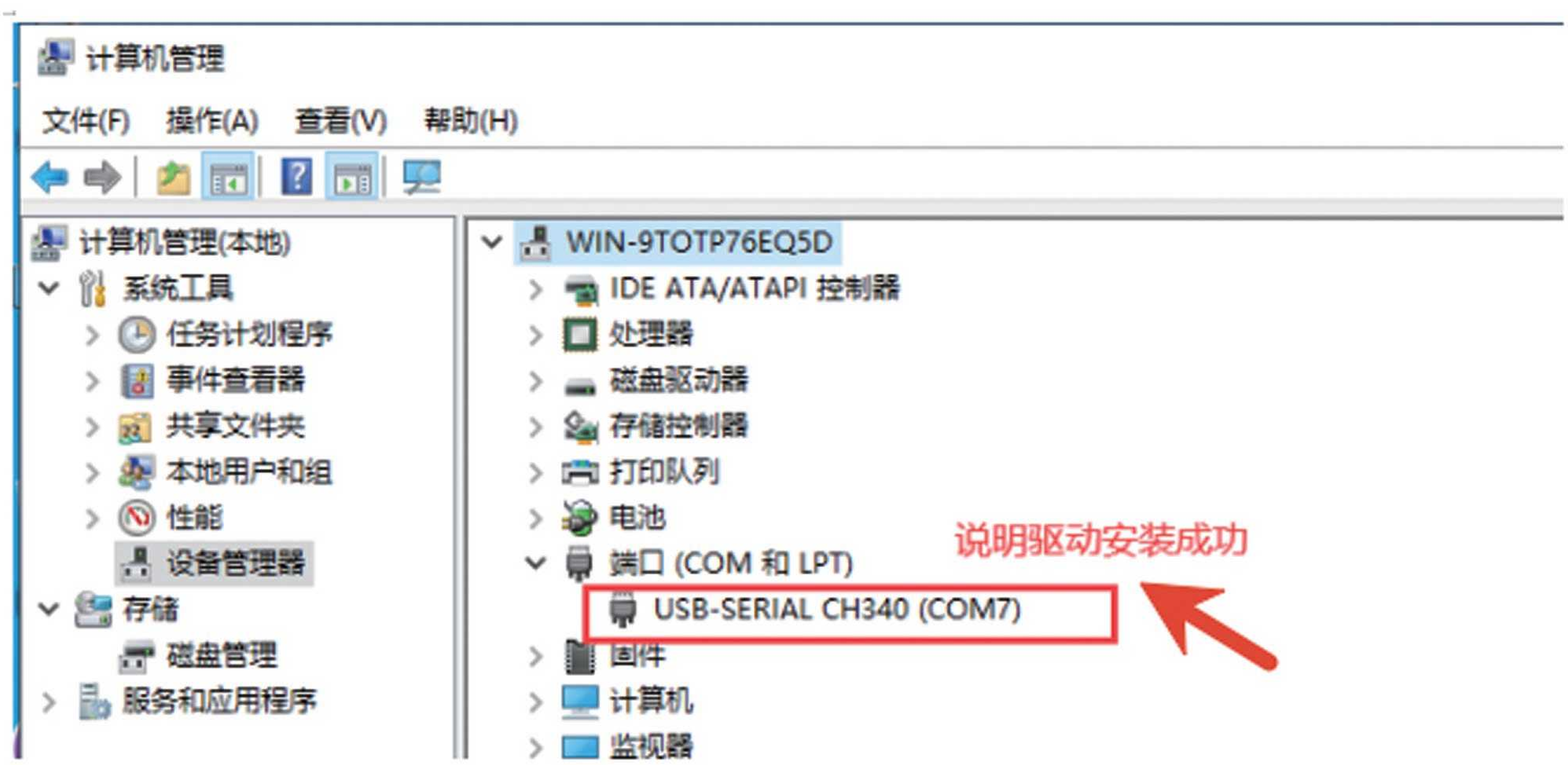
1、 Installation of serial port driver for development board

1. Open the provided 06 development board serial port driver CH343, double-click the driver, and click "Install" to start the installation

2. The following is the successful installation



3. Connect the development board to the computer using a USB cable



2、 Installation of Arduino development environment

1. Double click the Arduino installation package to install Arduino:

2. Code package path setting:

3. Code directory structure:

be careful:

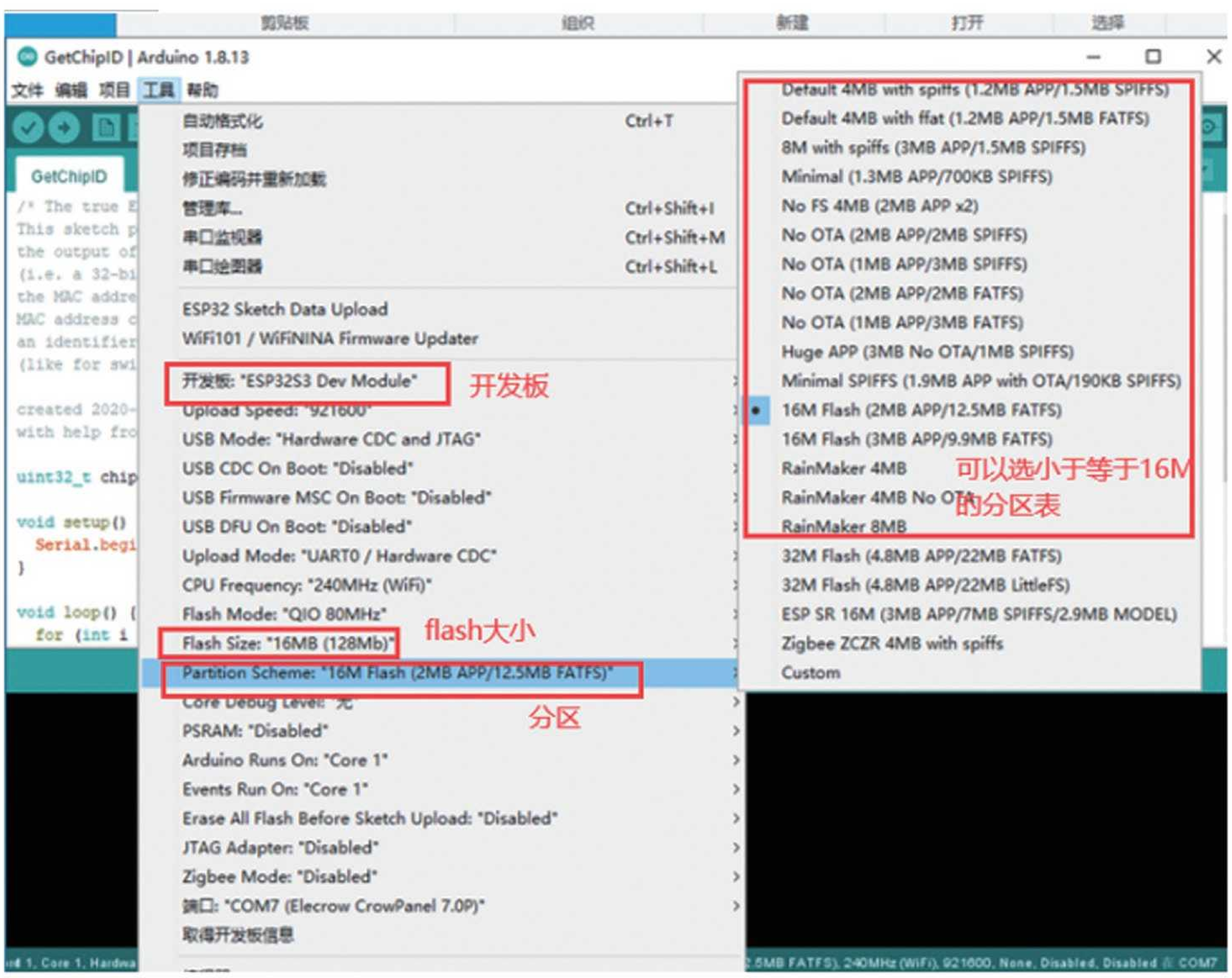
1) Be sure to manually create the hardware directory

2) Extract the code directly to the hardwire directory

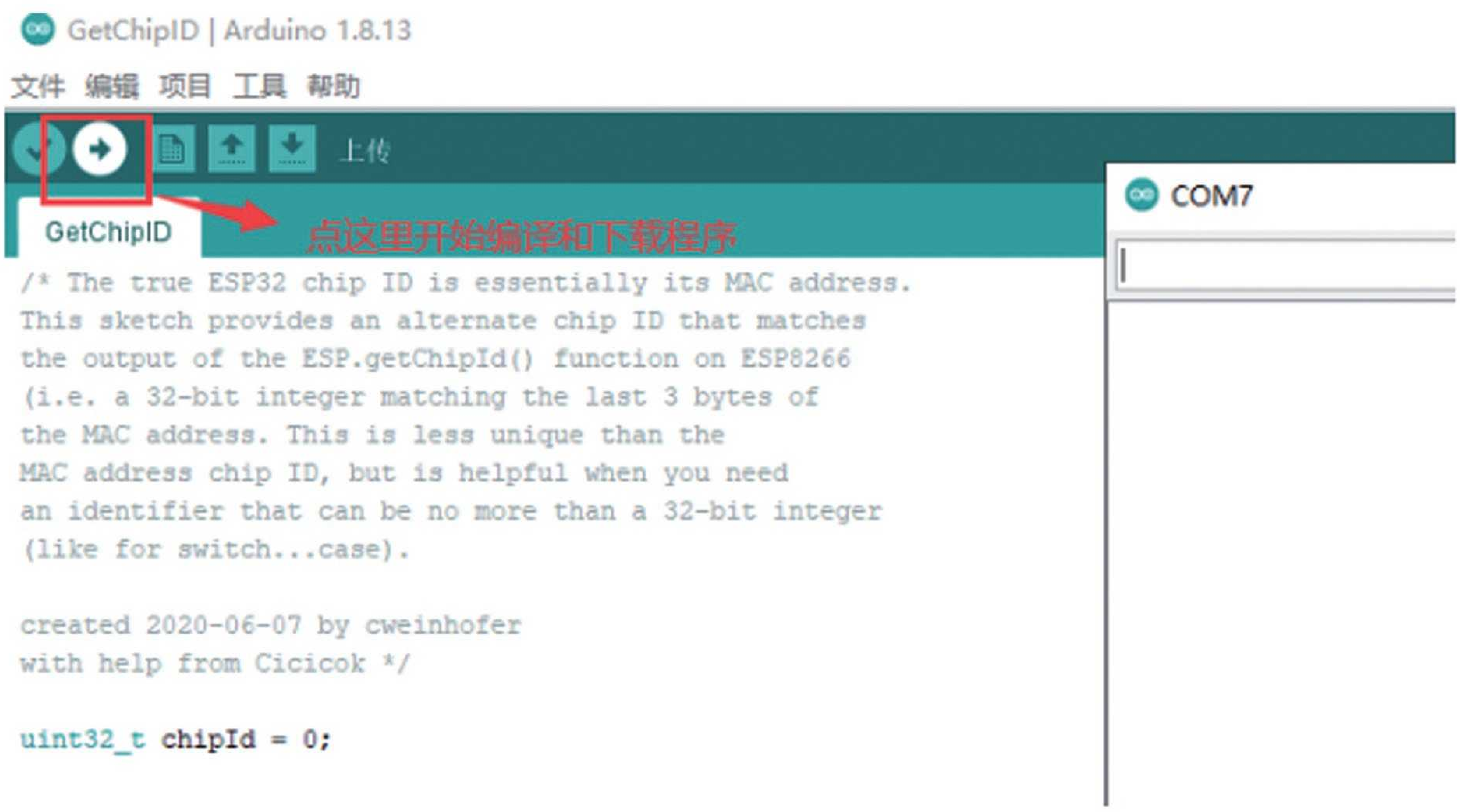
3) There are two ESP32 directories

After setting the path, be sure to restart Arduino!!!!

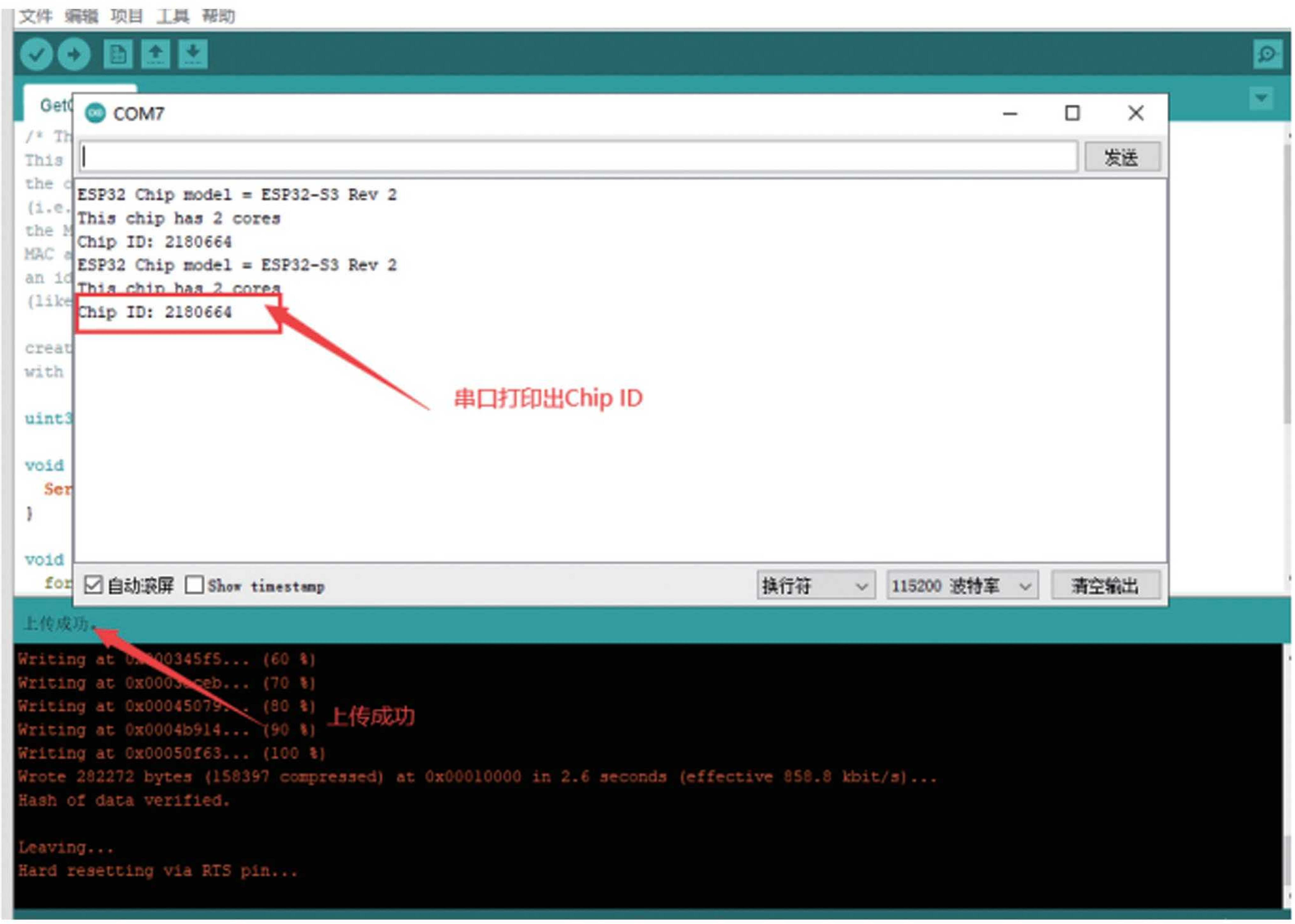
4. Open a routine below, select the development board, serial port, and partition for verification, and then click "Upload" to verify that the environment has been installed:



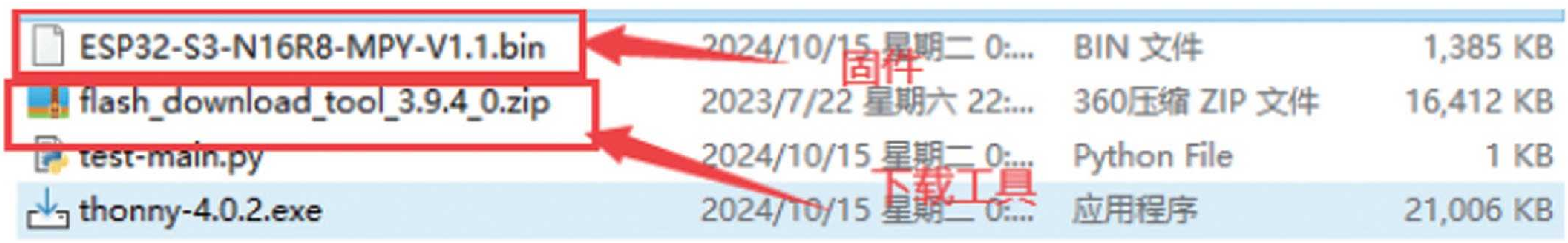
5. Download verification:



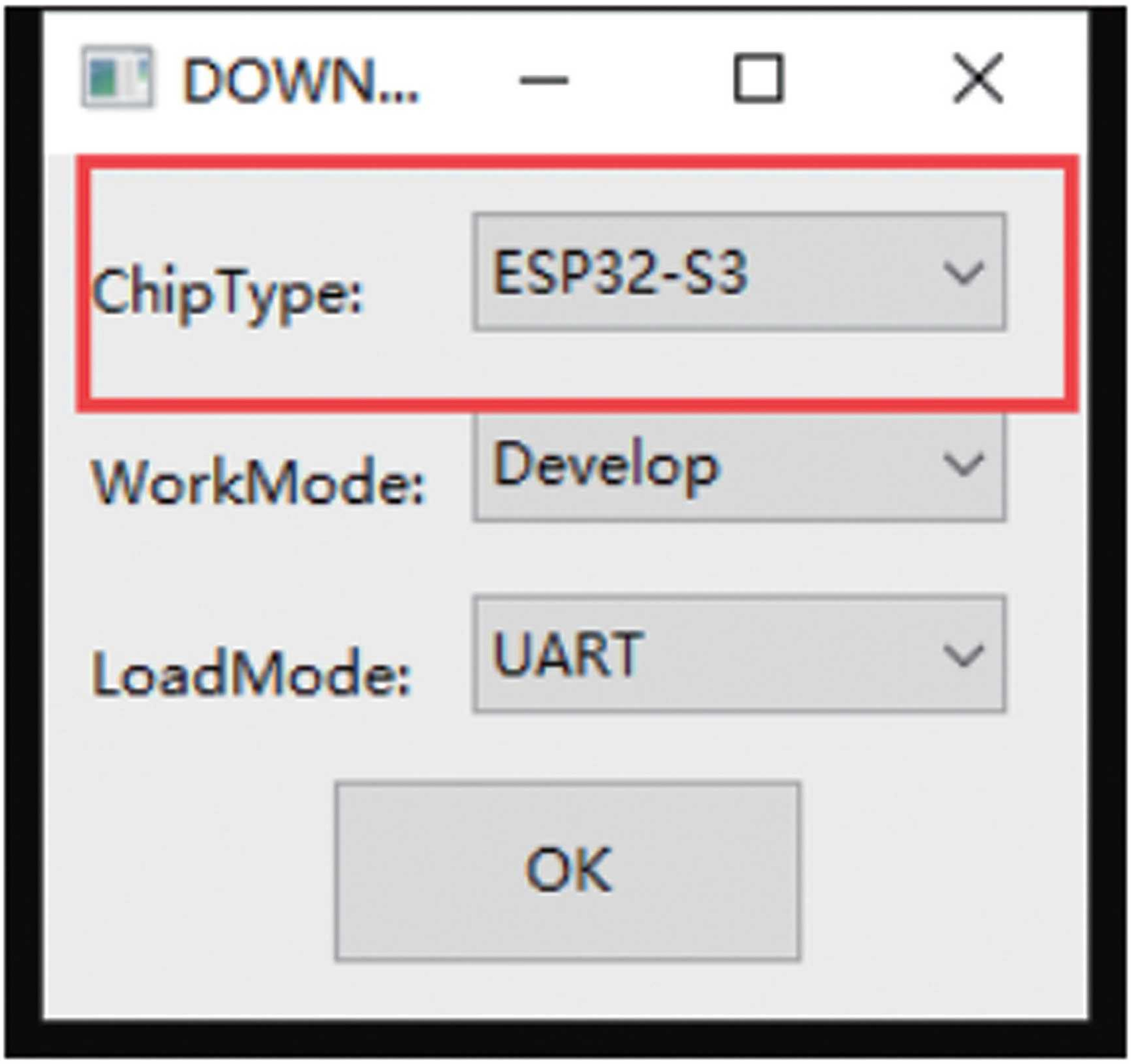
6. It can be seen through the "serial monitor" that chipID is being played, indicating that the environment is ready:



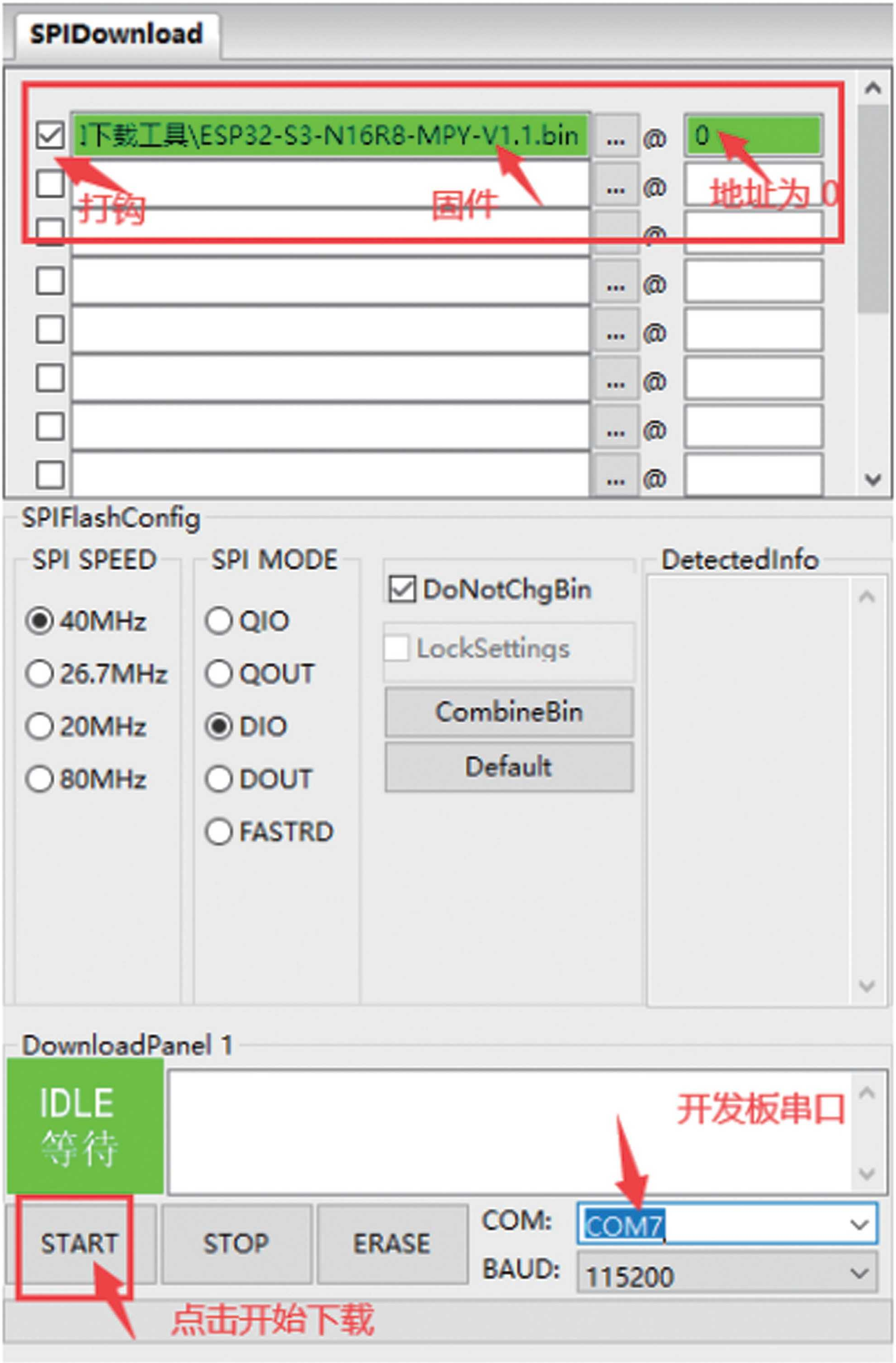
3、 Installation of Micropython Development Environment



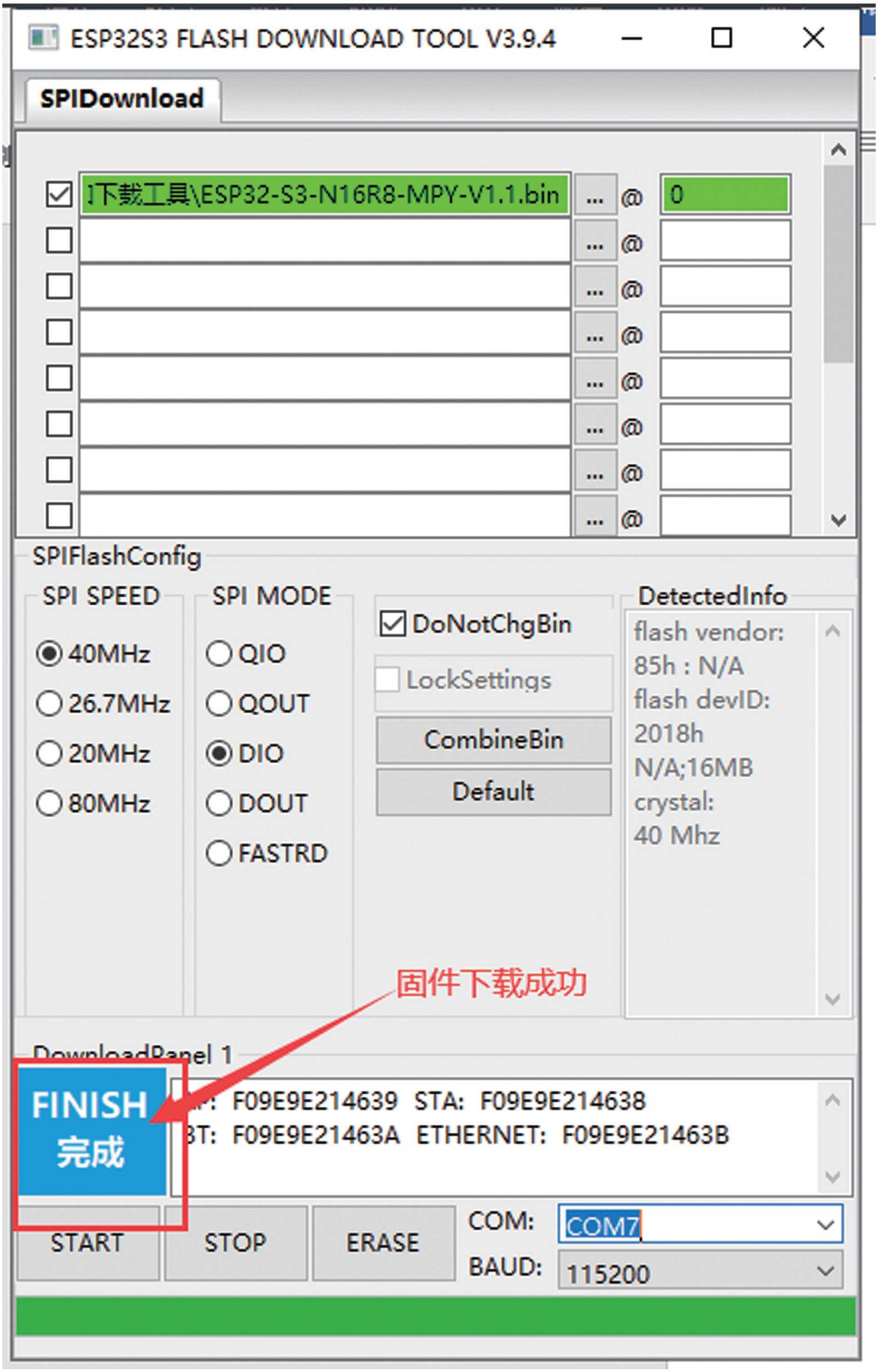
1. Extract the micropython download toolkit, double-click to open the micropython firmware download tool, and select "ESP32-S3"



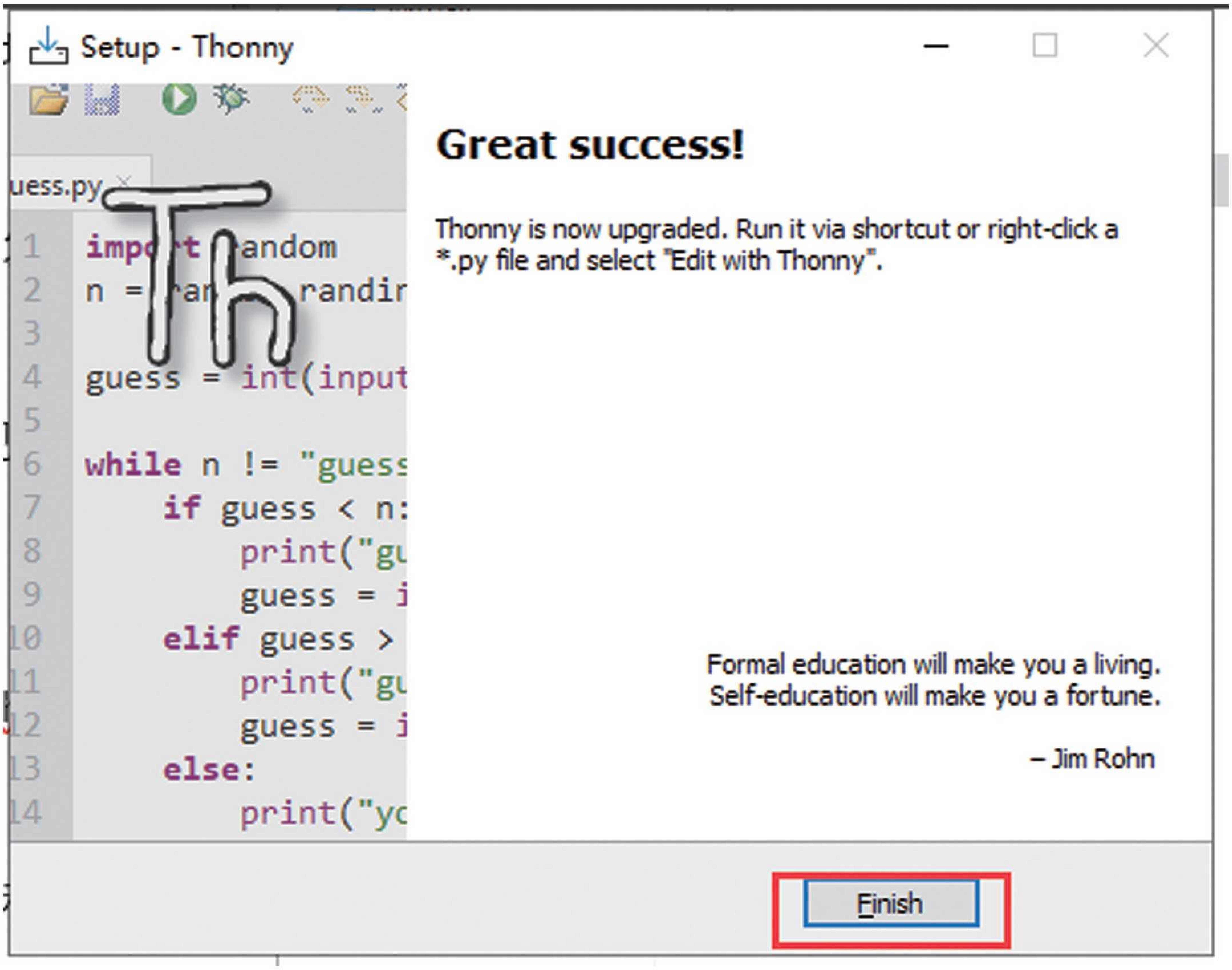
2. Select firmware to start downloading:



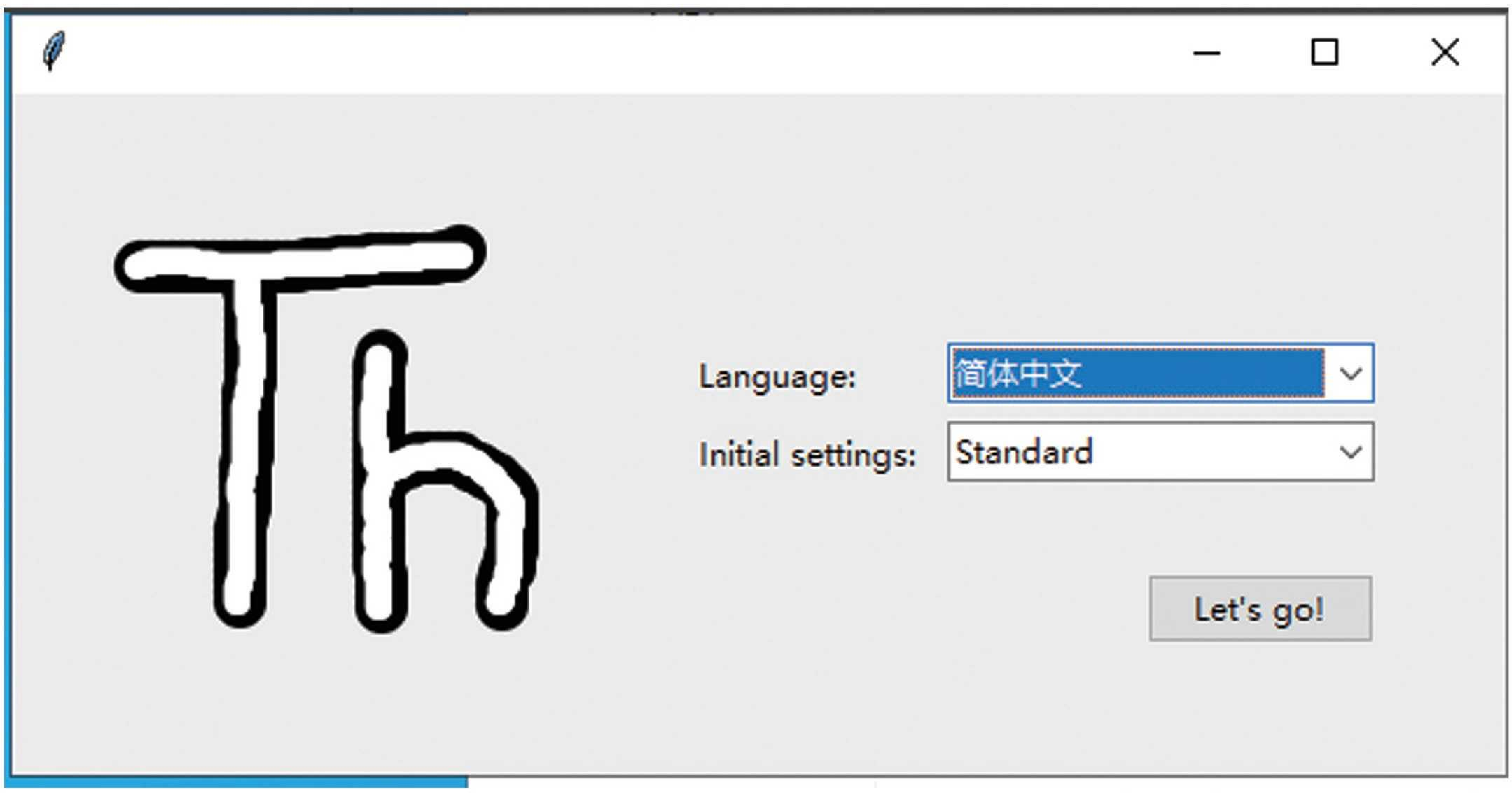
3. Click "START" to start downloading:



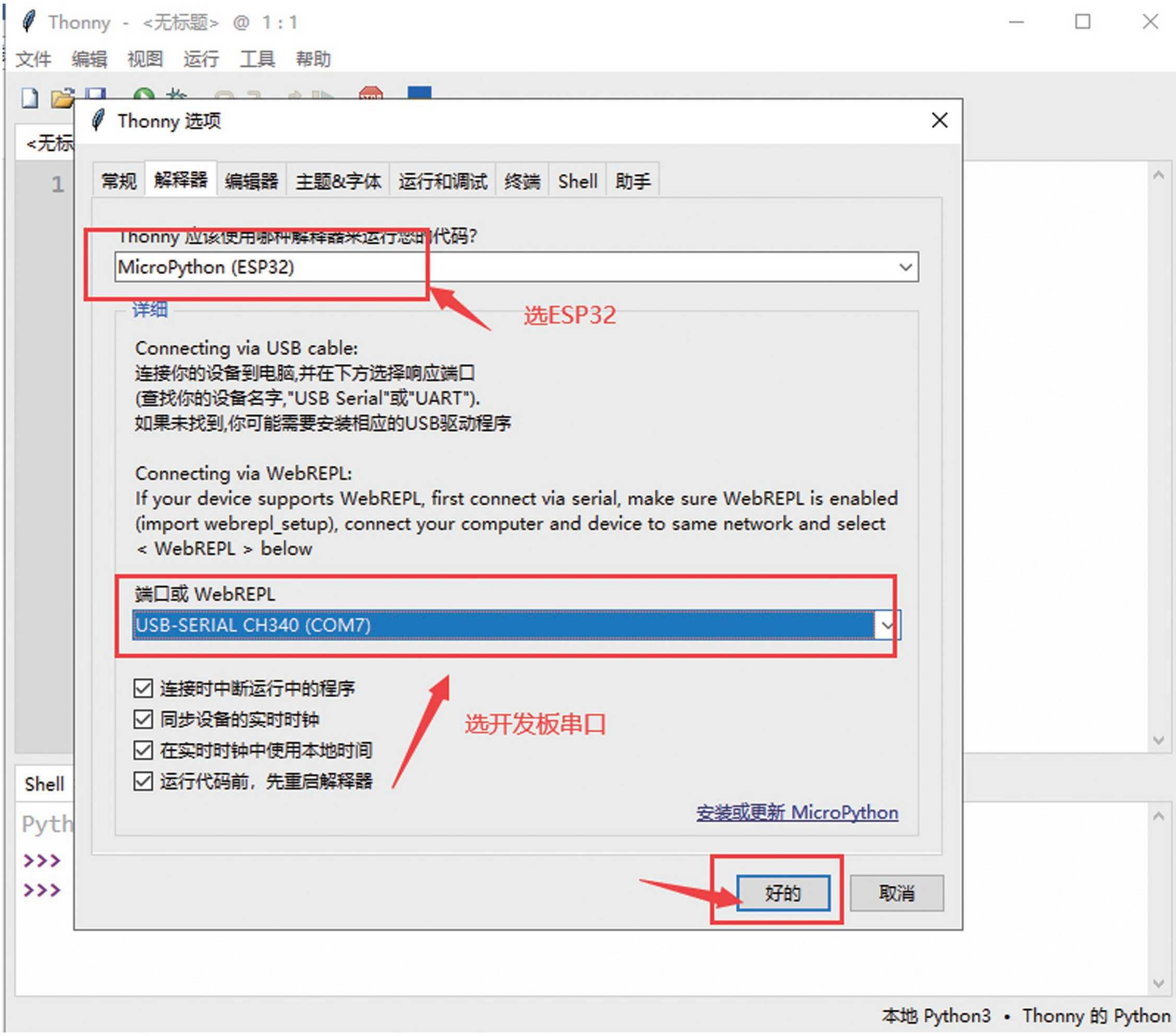
4. Double click to install thongy-4.0.2.exe, the next step is to complete the installation:



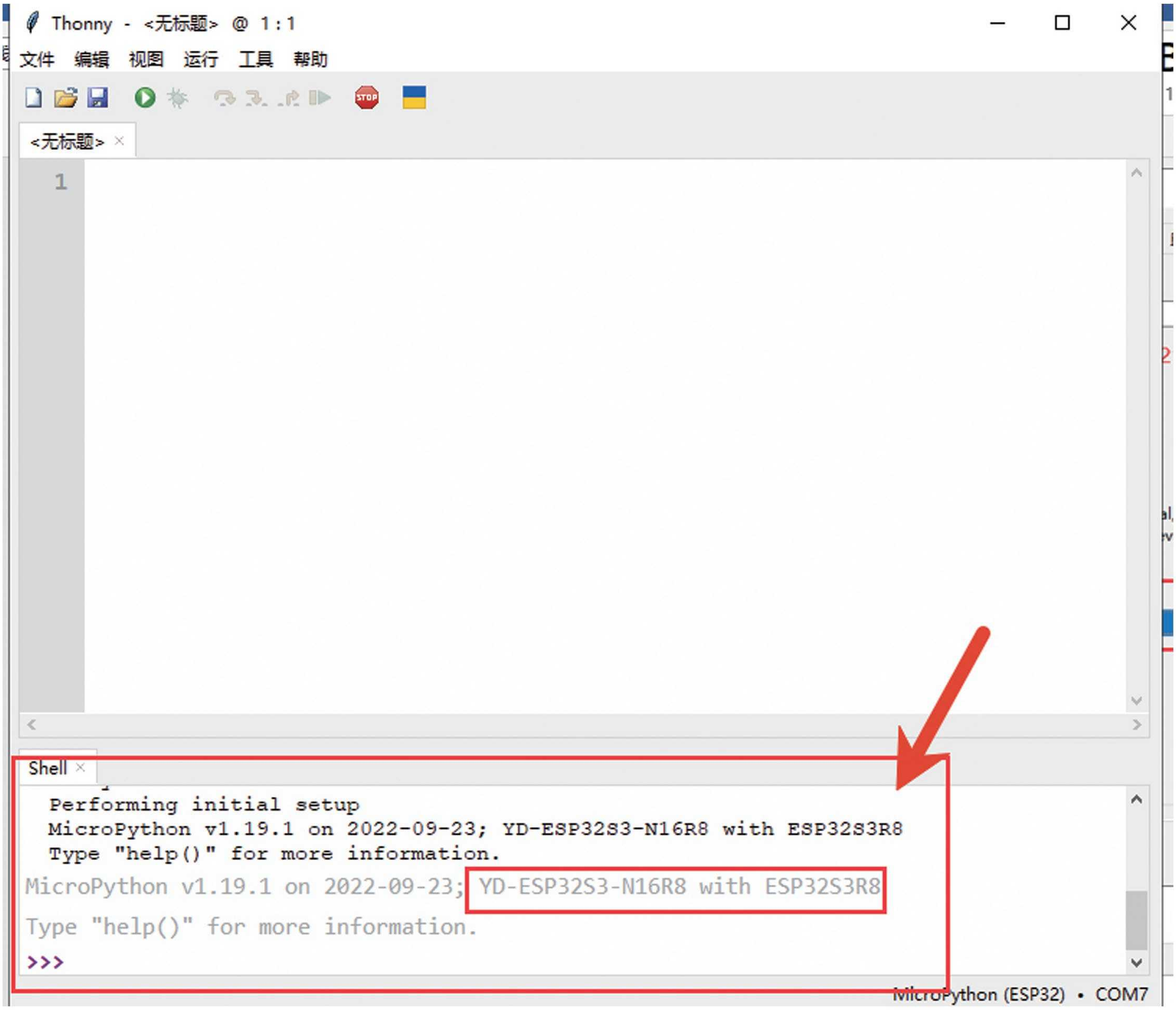
5. Open Python for environment validation, select the language option:



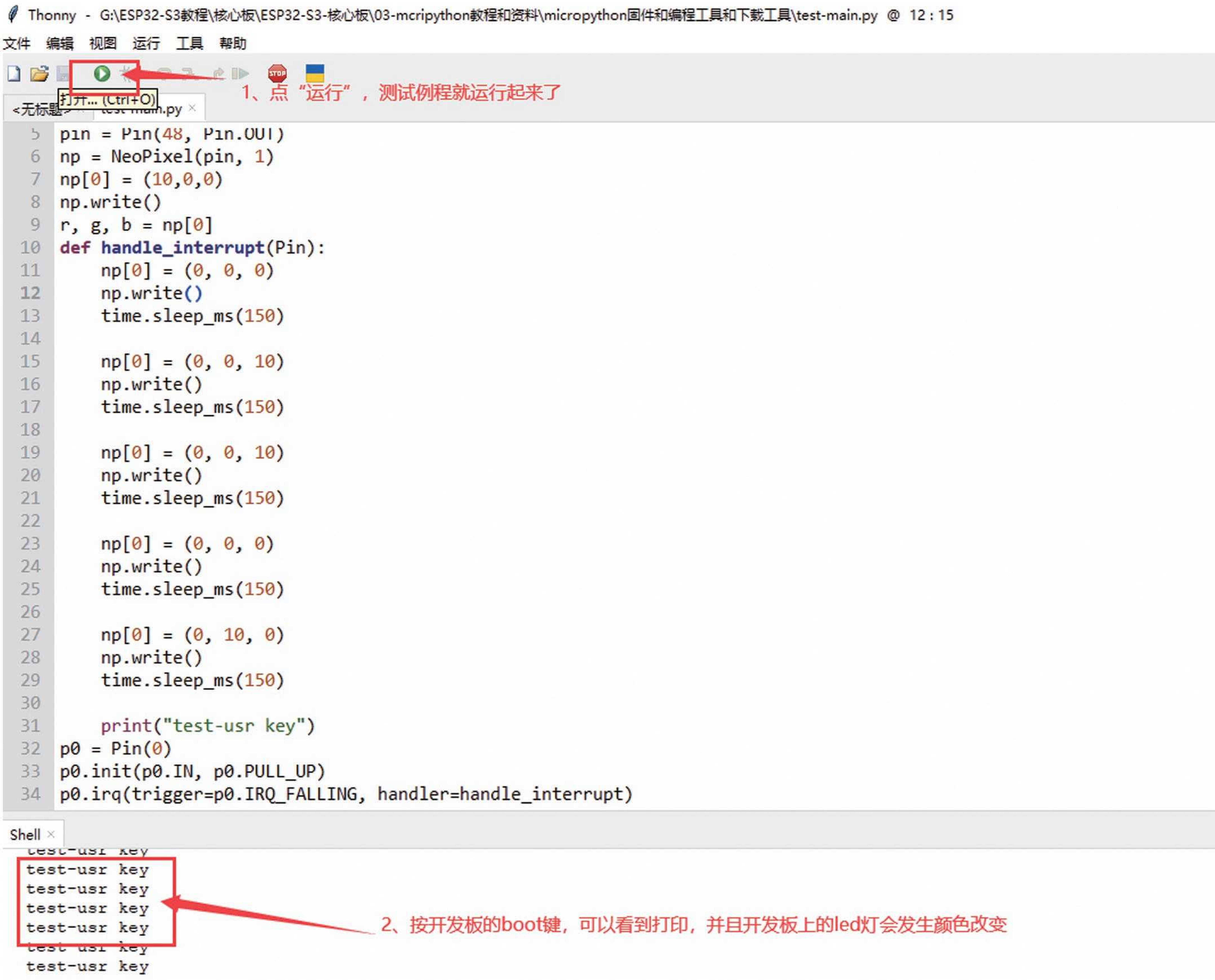
6. Select configuration, "Run" ->"Configure Interpreter"



Press the RST key on the development board, and you will see that Python has already connected to the board,



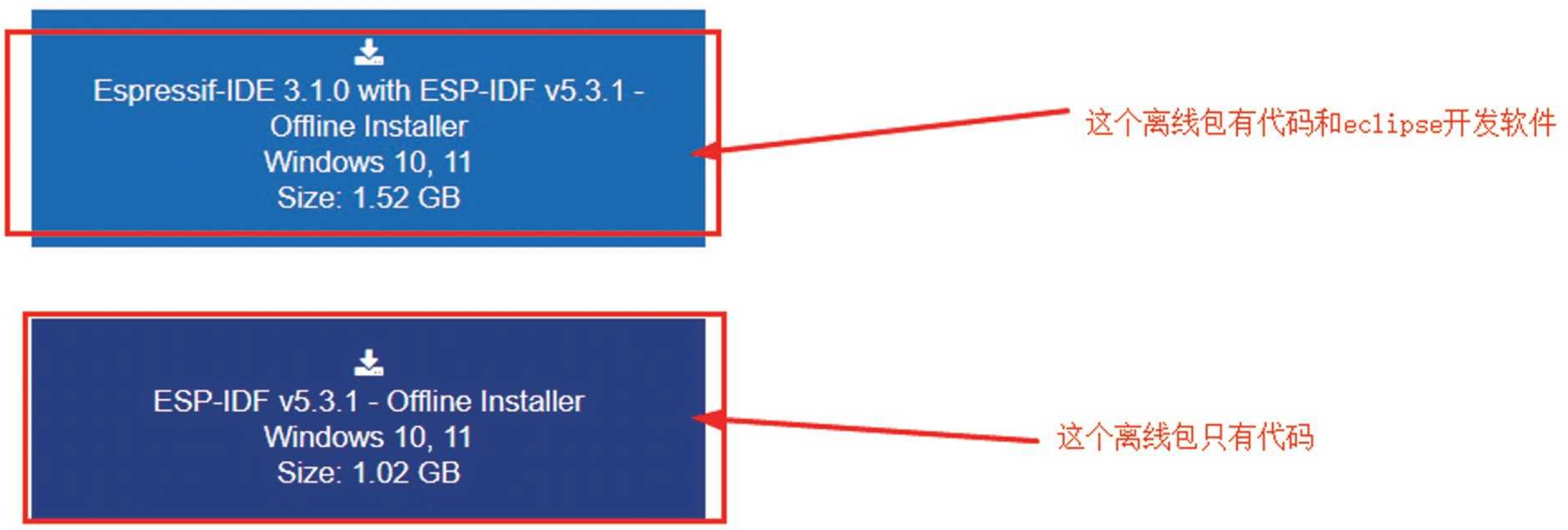
7. Open a test history test main.cy to conduct testing



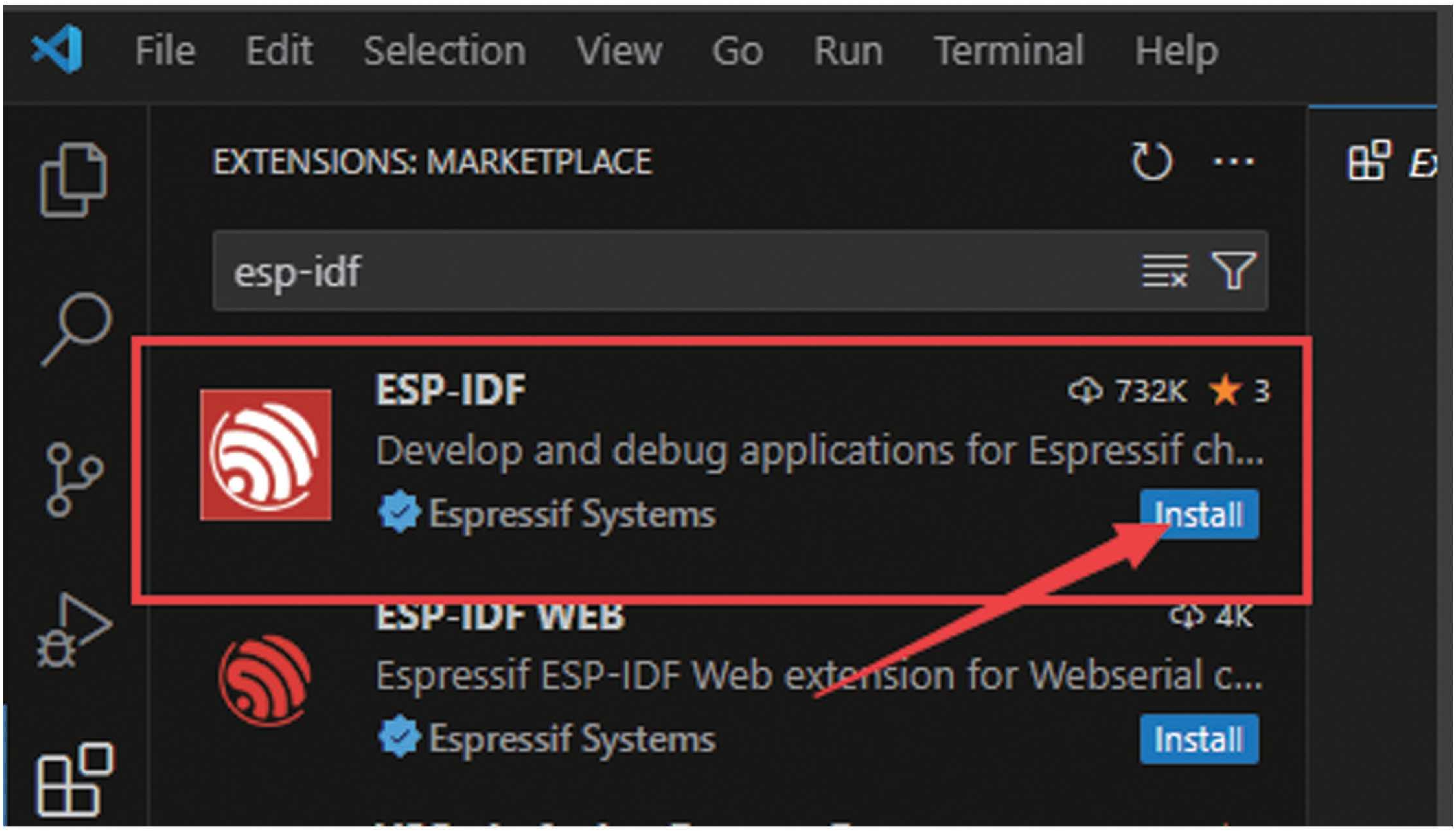
At this point, the micropython development environment has been set up.

4、 Installation of Idf development environment

1. Idf offline package download address: https://dl.espressif.cn/dl/esp-id-f/?idf=4.4 You can download offline packages from this address



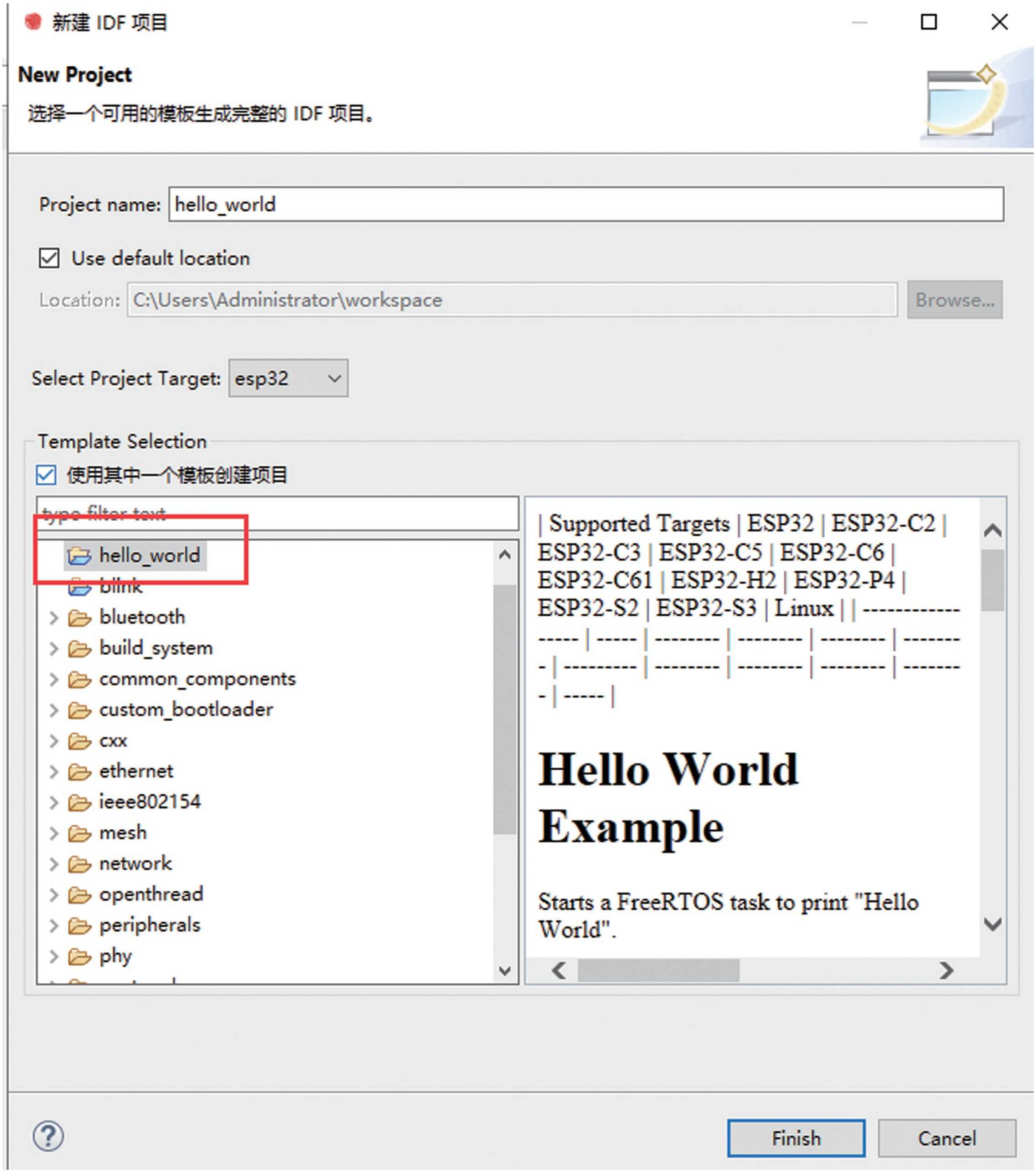
2. Install ESP-IDF plugin on VSCode



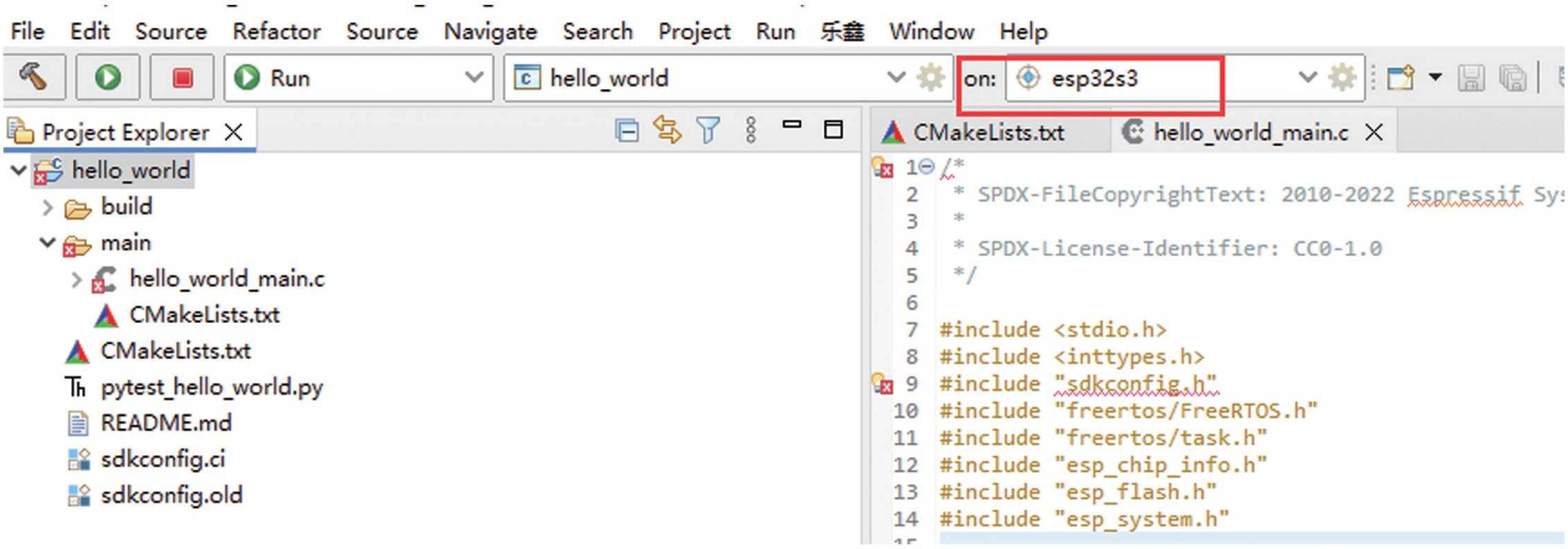
Next, let's watch the video tutorial on how to configure and solve the problems encountered.

1) Developing with Eclipse

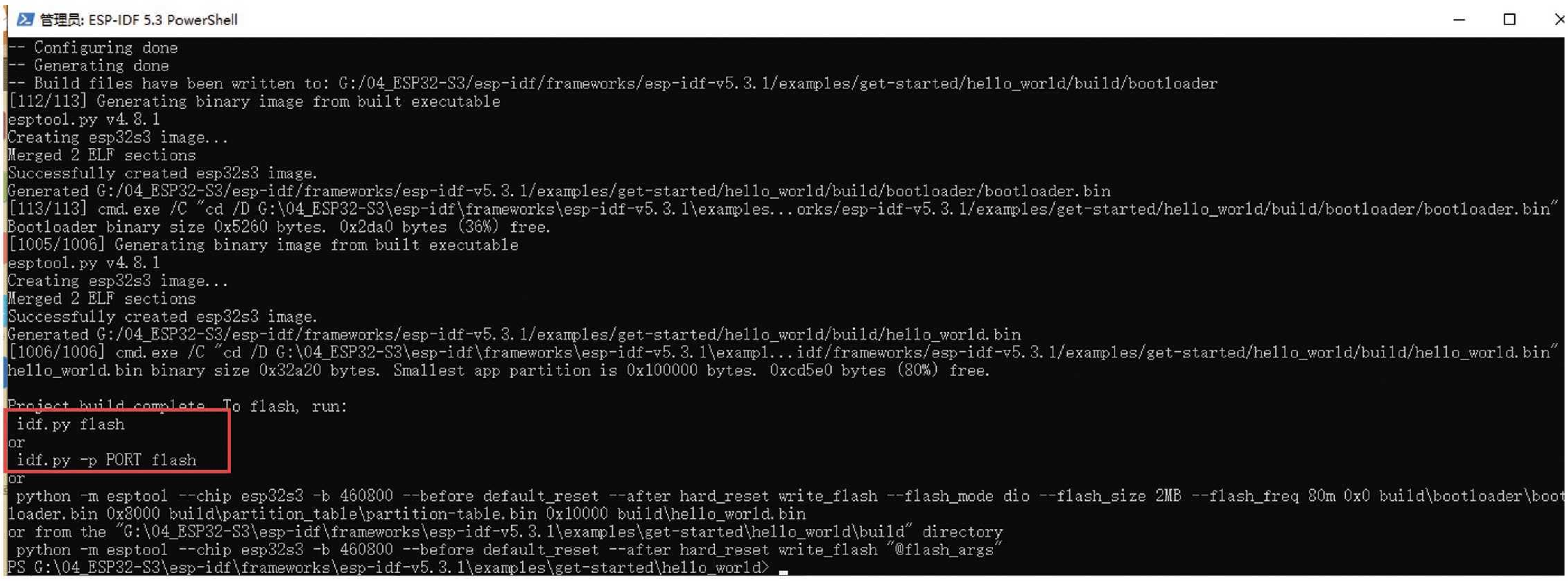
If the installation of the Python package is slow because it is installed using foreign Python sources by default, you can search online to set up domestic Python sources.



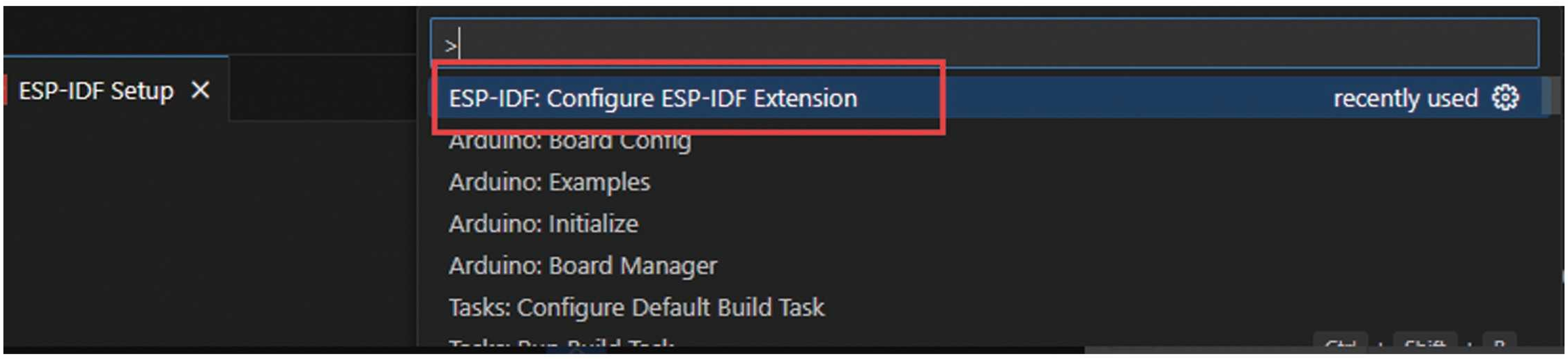
Set the compilation target to esp32s3

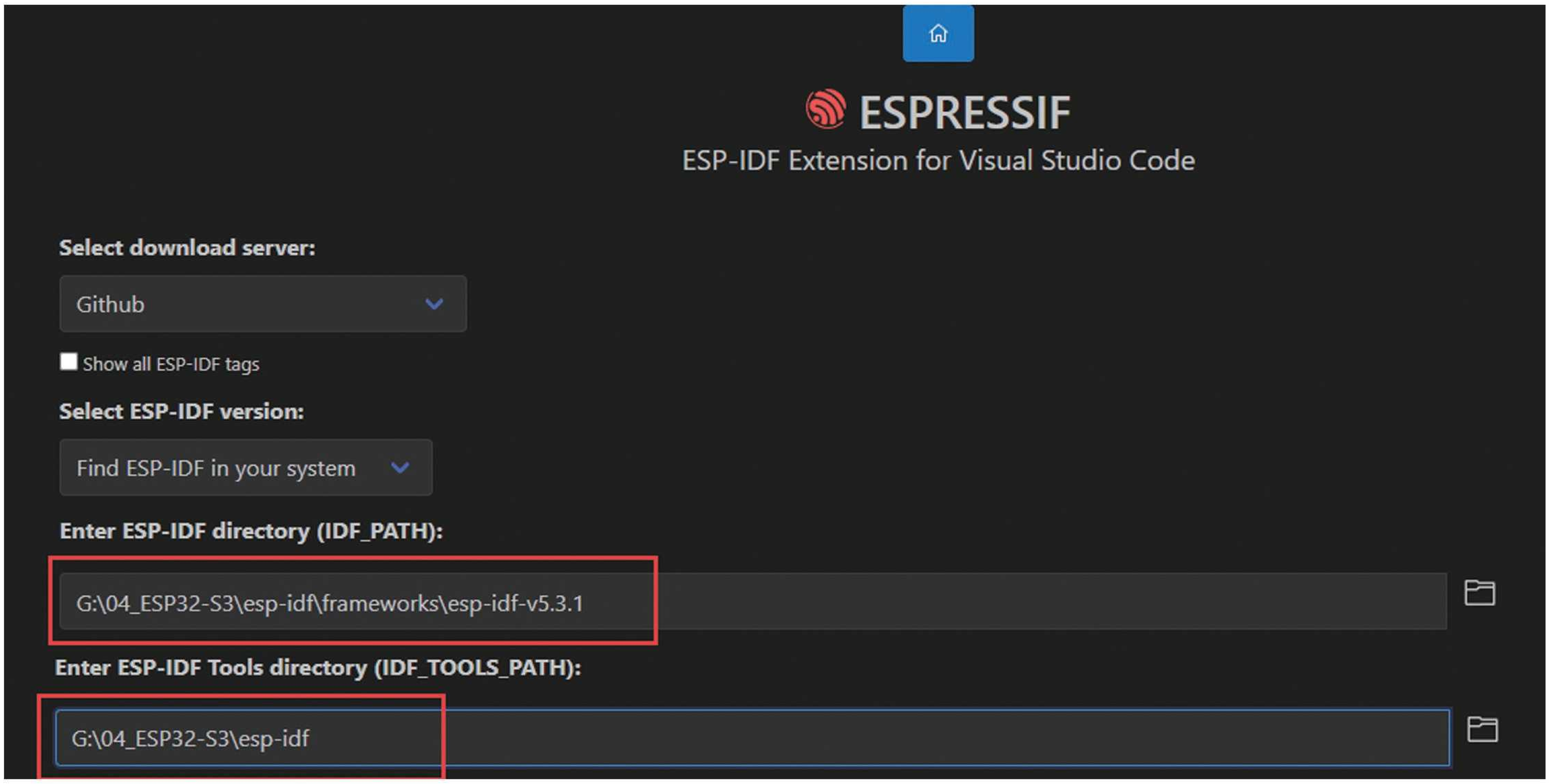


2) Develop using the Windows command line

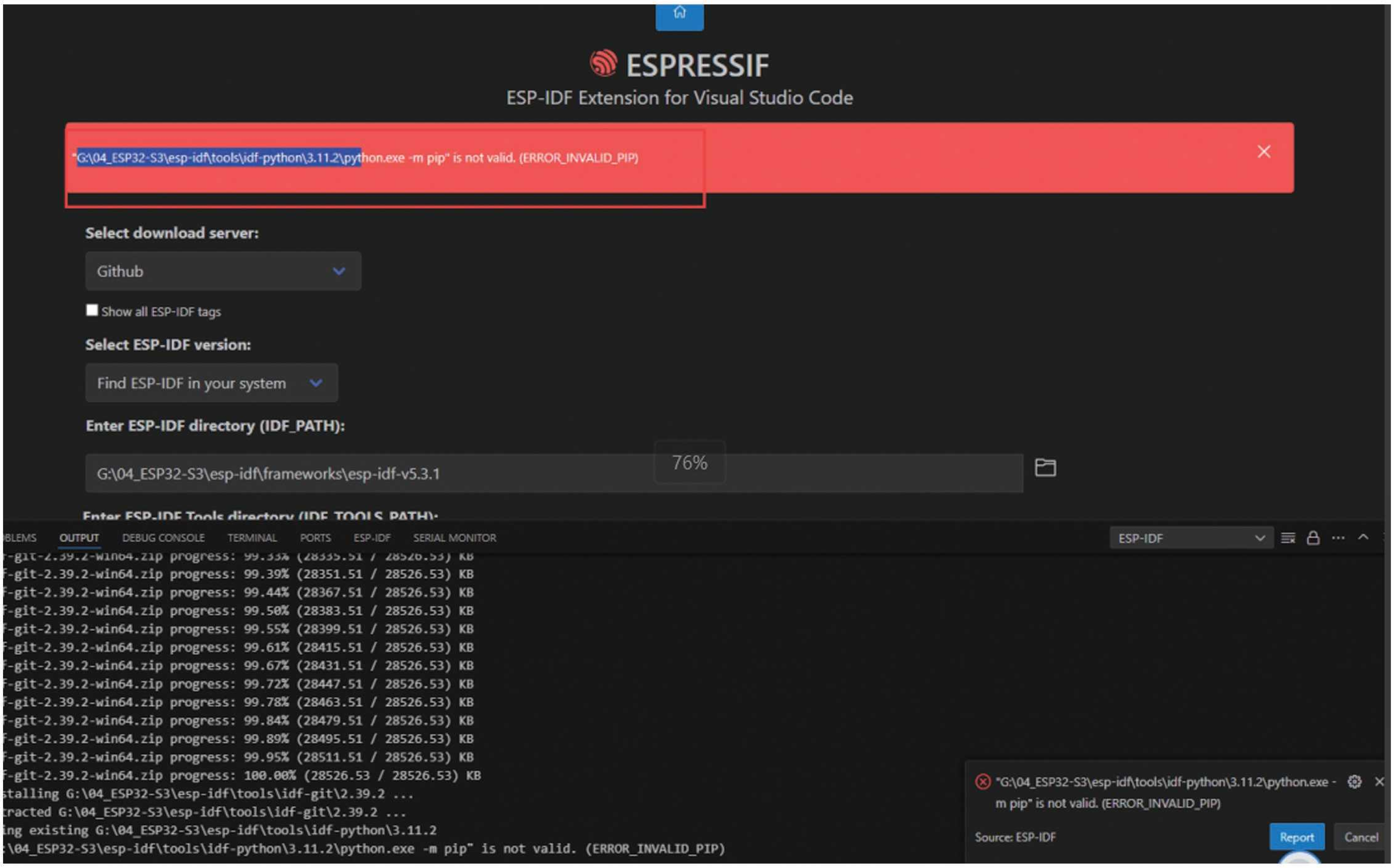


3) Developing with VSCode (highly recommended)

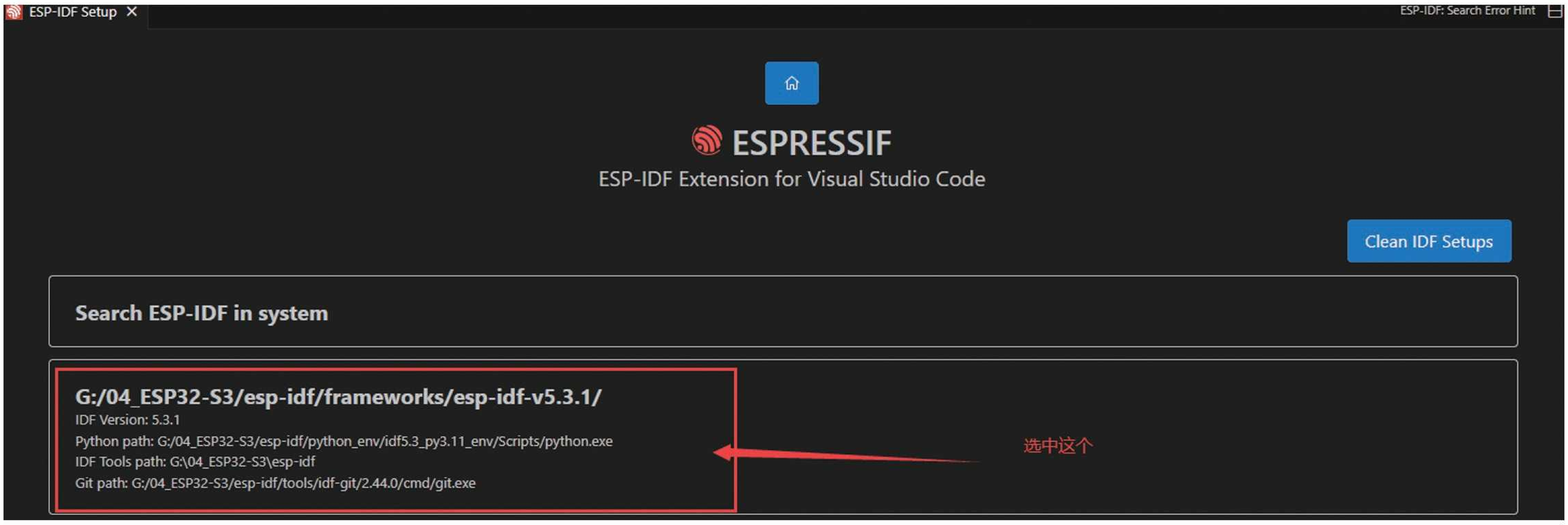


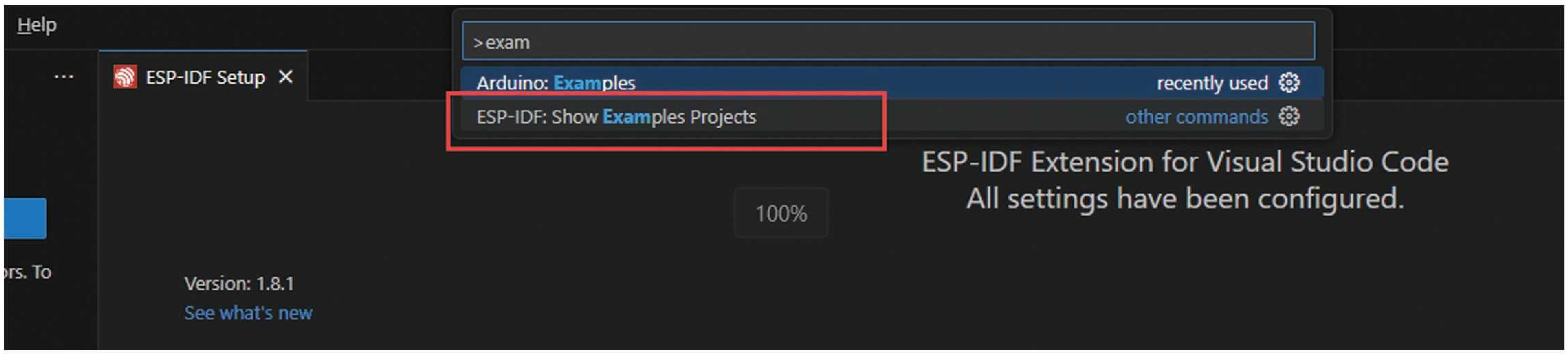


The following error occurred:

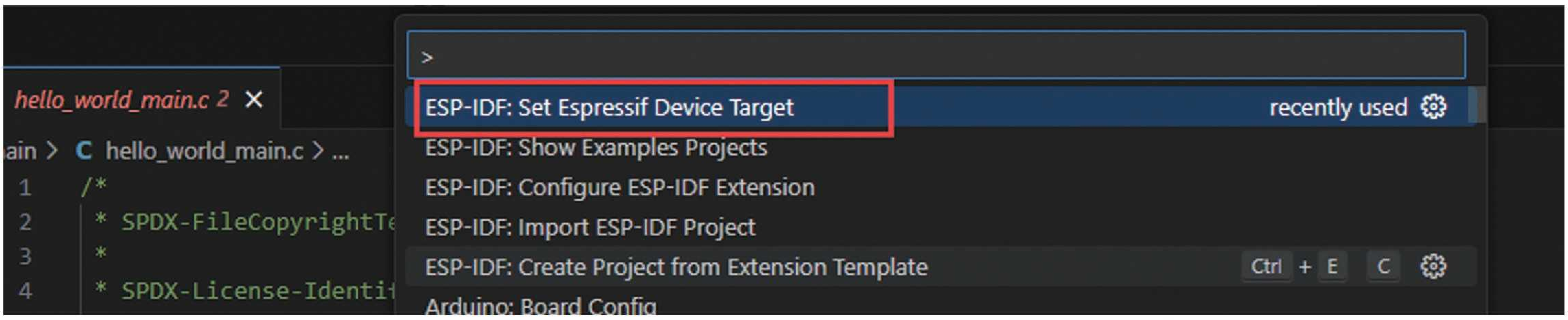


Solution: Restart VSCode and select the configuration again to solve the problem.

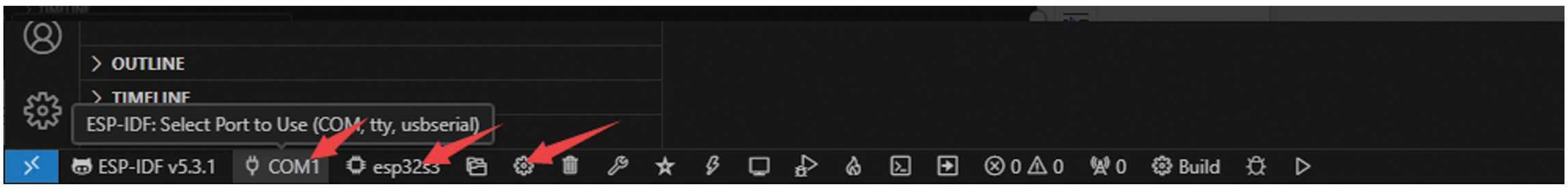




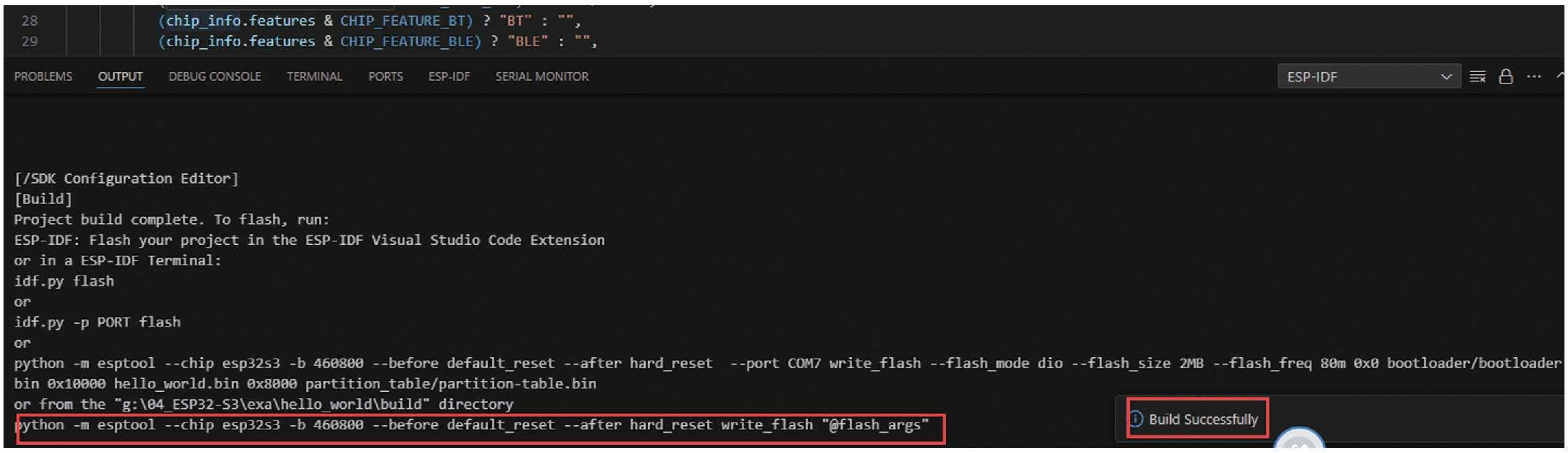
Configure development board:



There is a quick configuration tool in the bottom left corner:



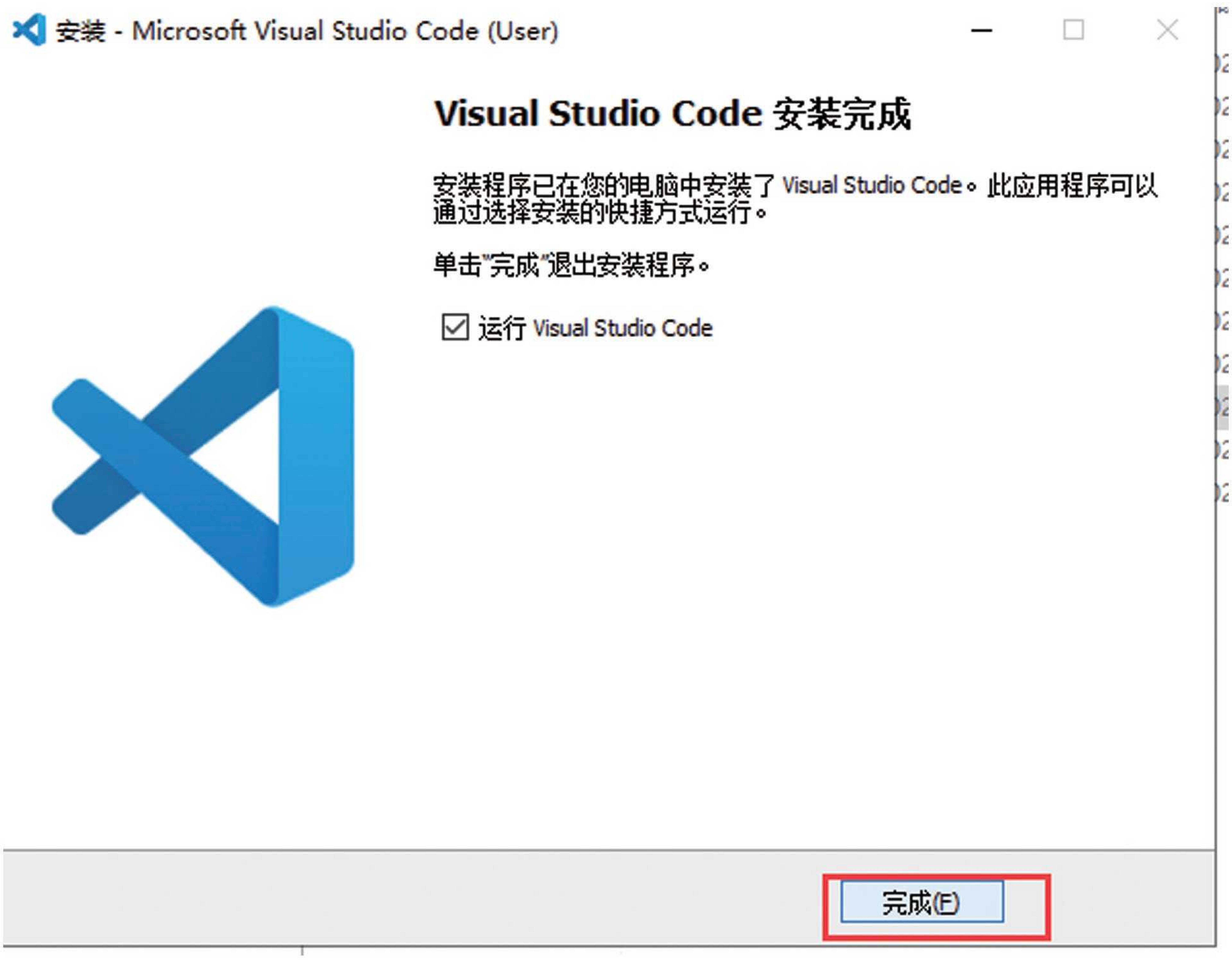
Compilation completed:



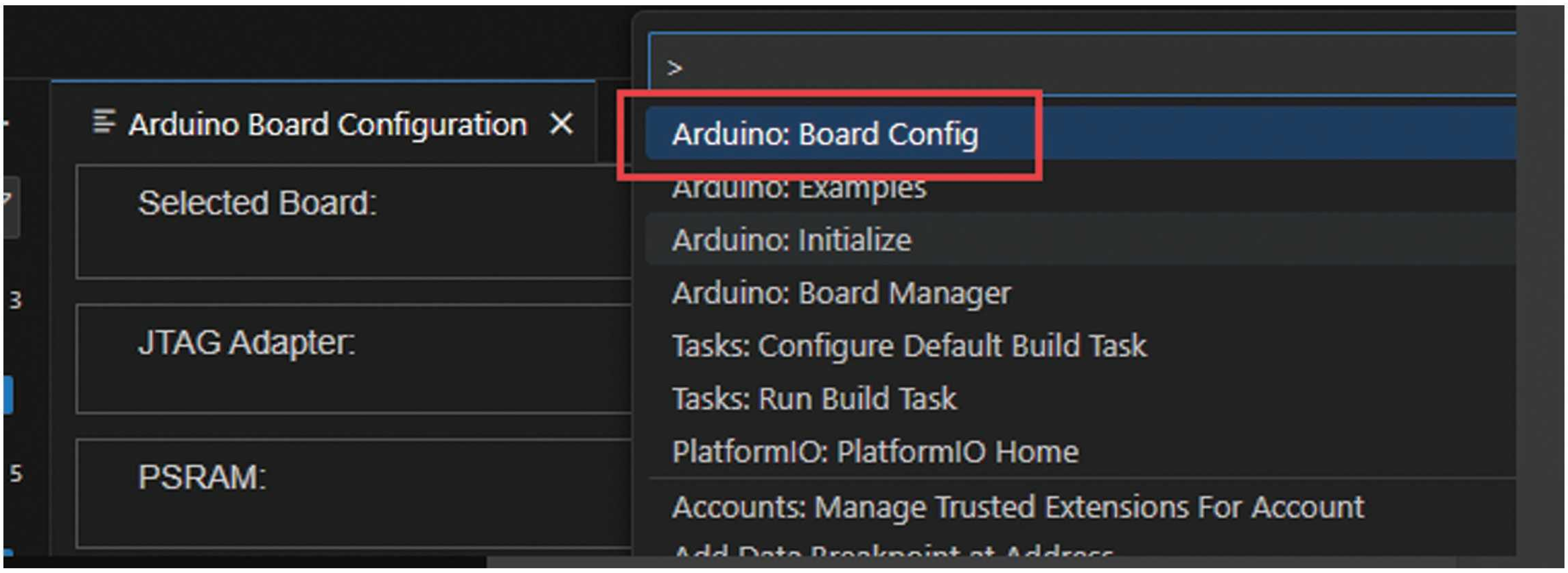
5、 Vscode development environment setup

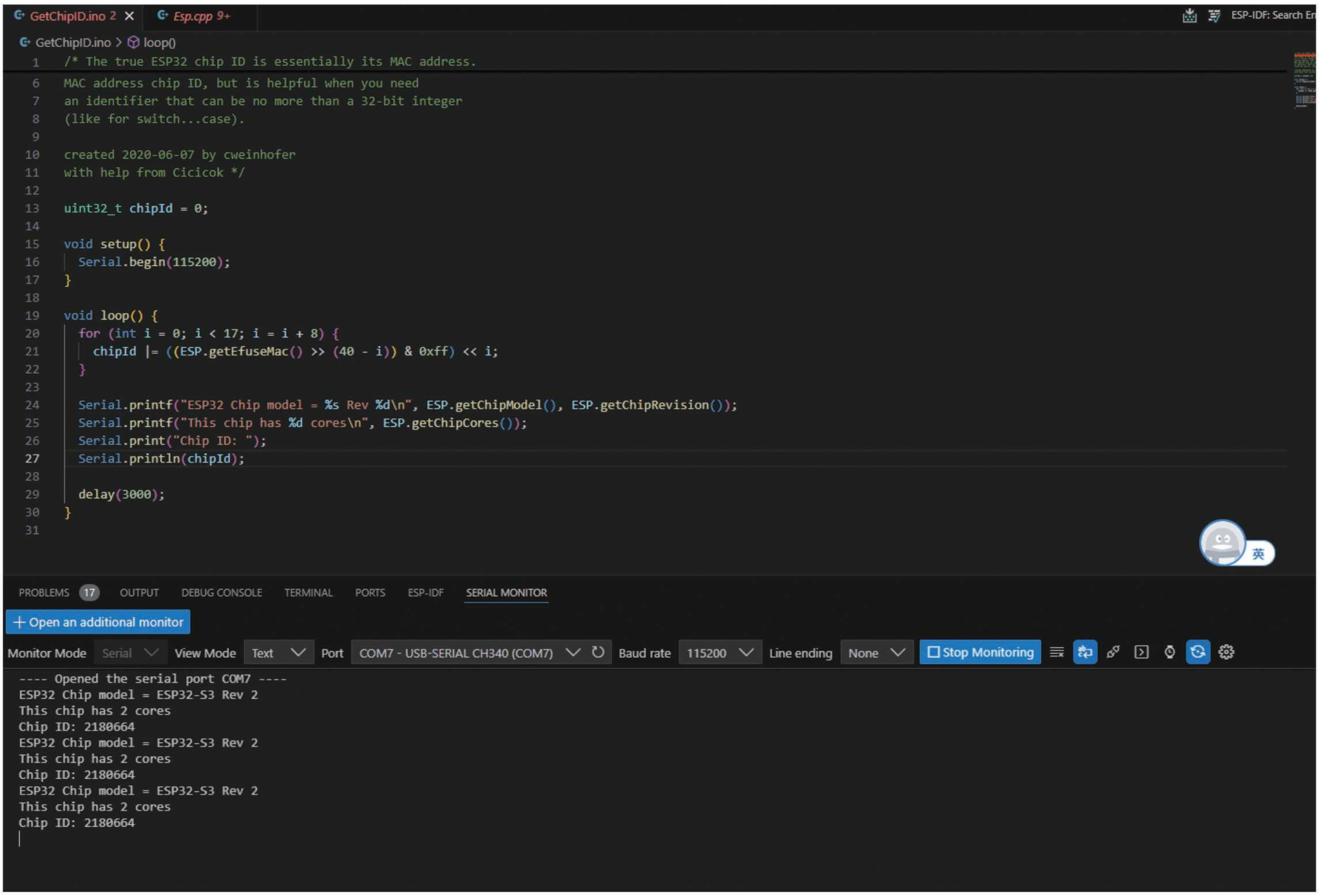
After installing the Arduino or IDF development environment, VSCode will only work

1) Double click the VSCode installation package to install VSCode,

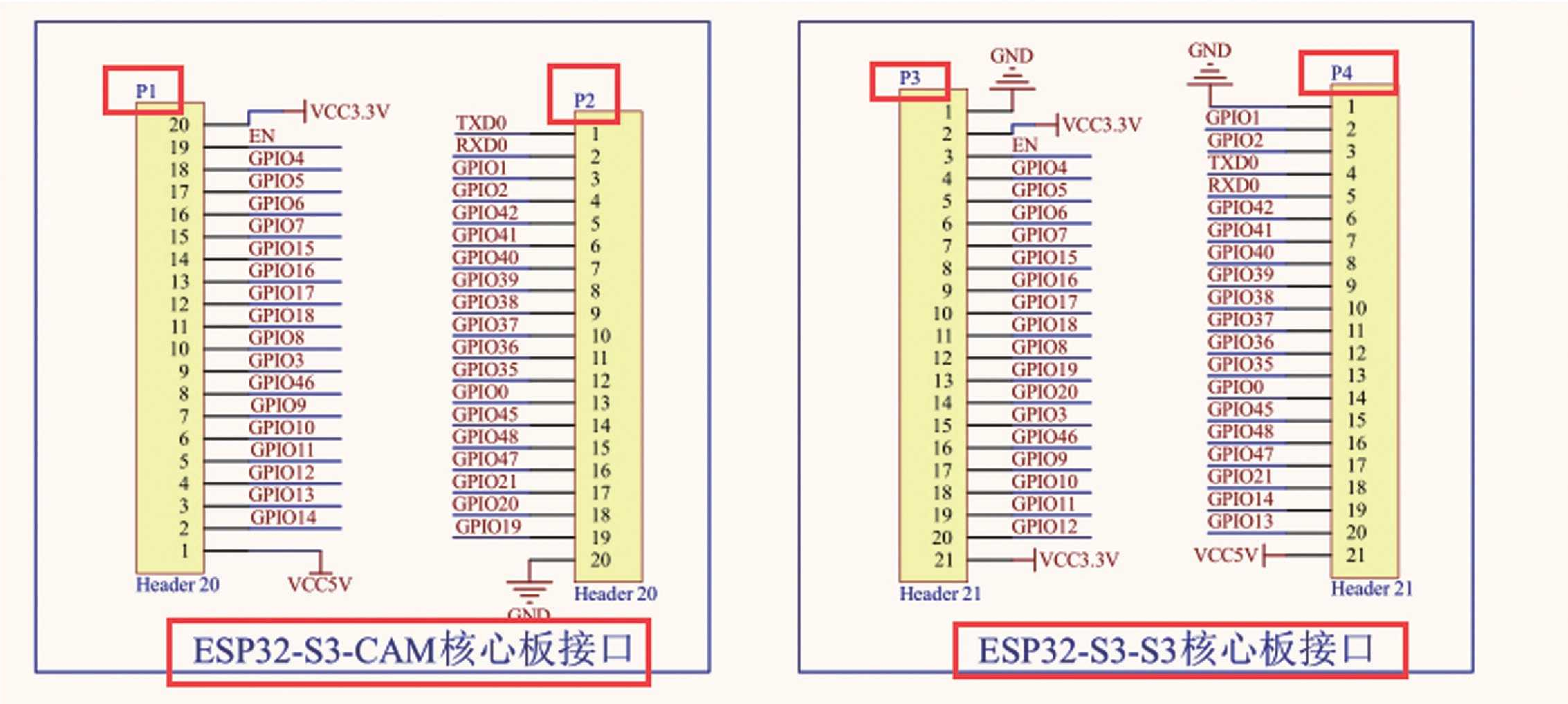


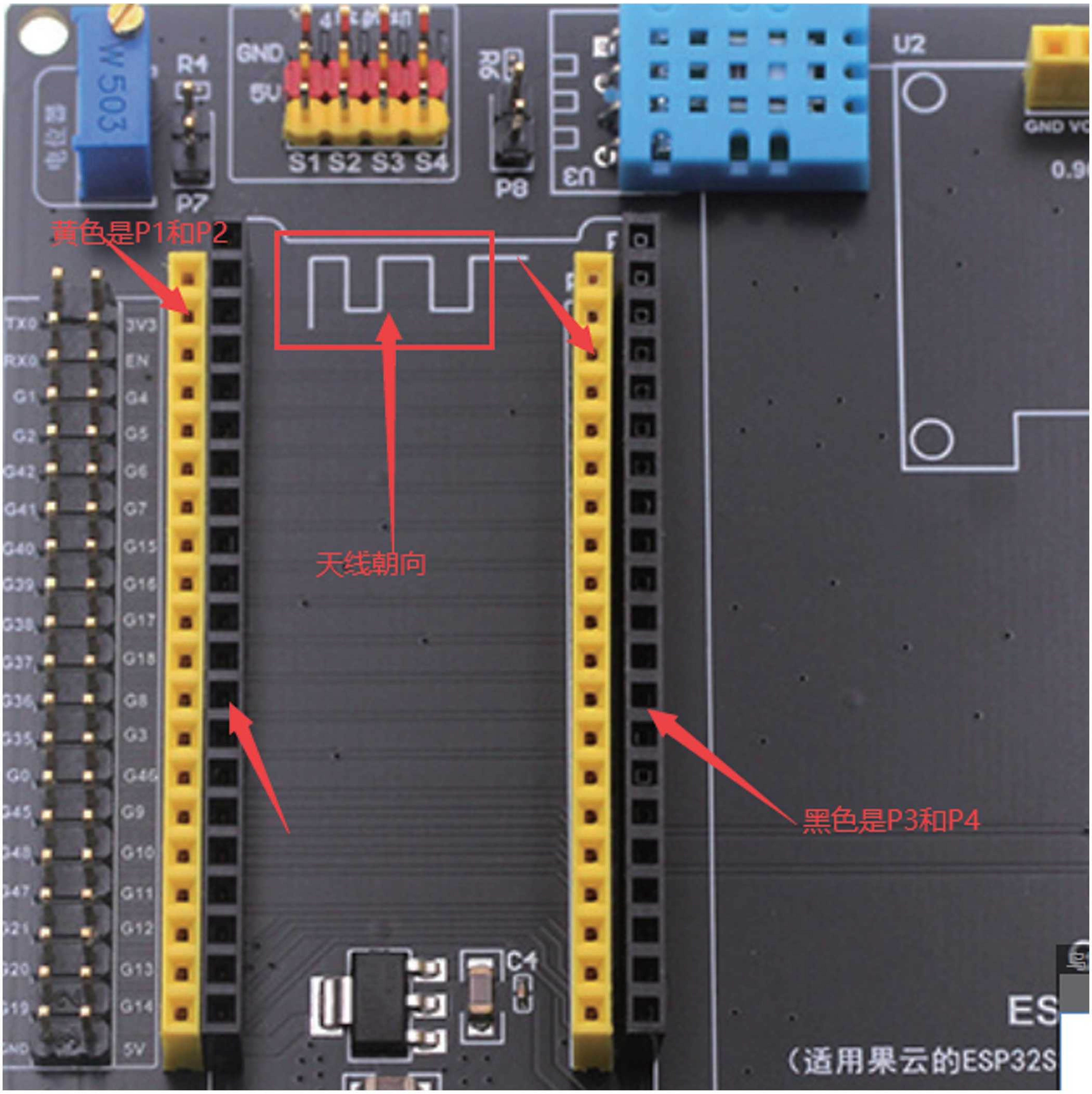
2) In the plugin market, install Arduino plugin or IDF plugin





Expansion board plug-in board instructions:

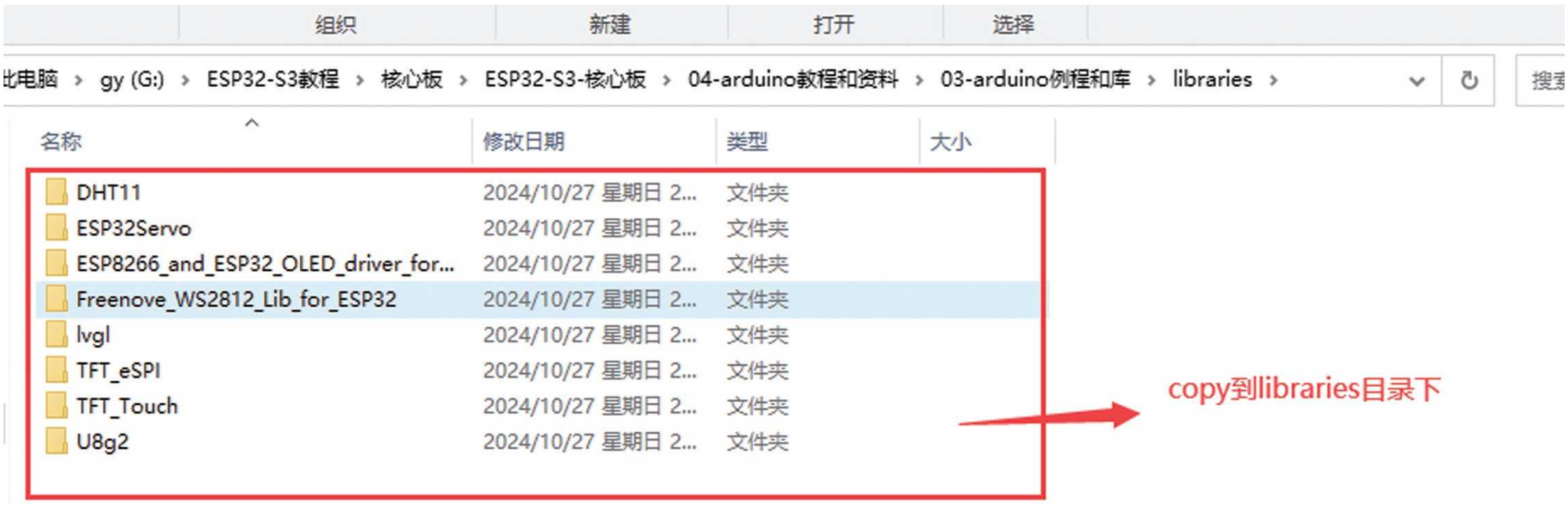


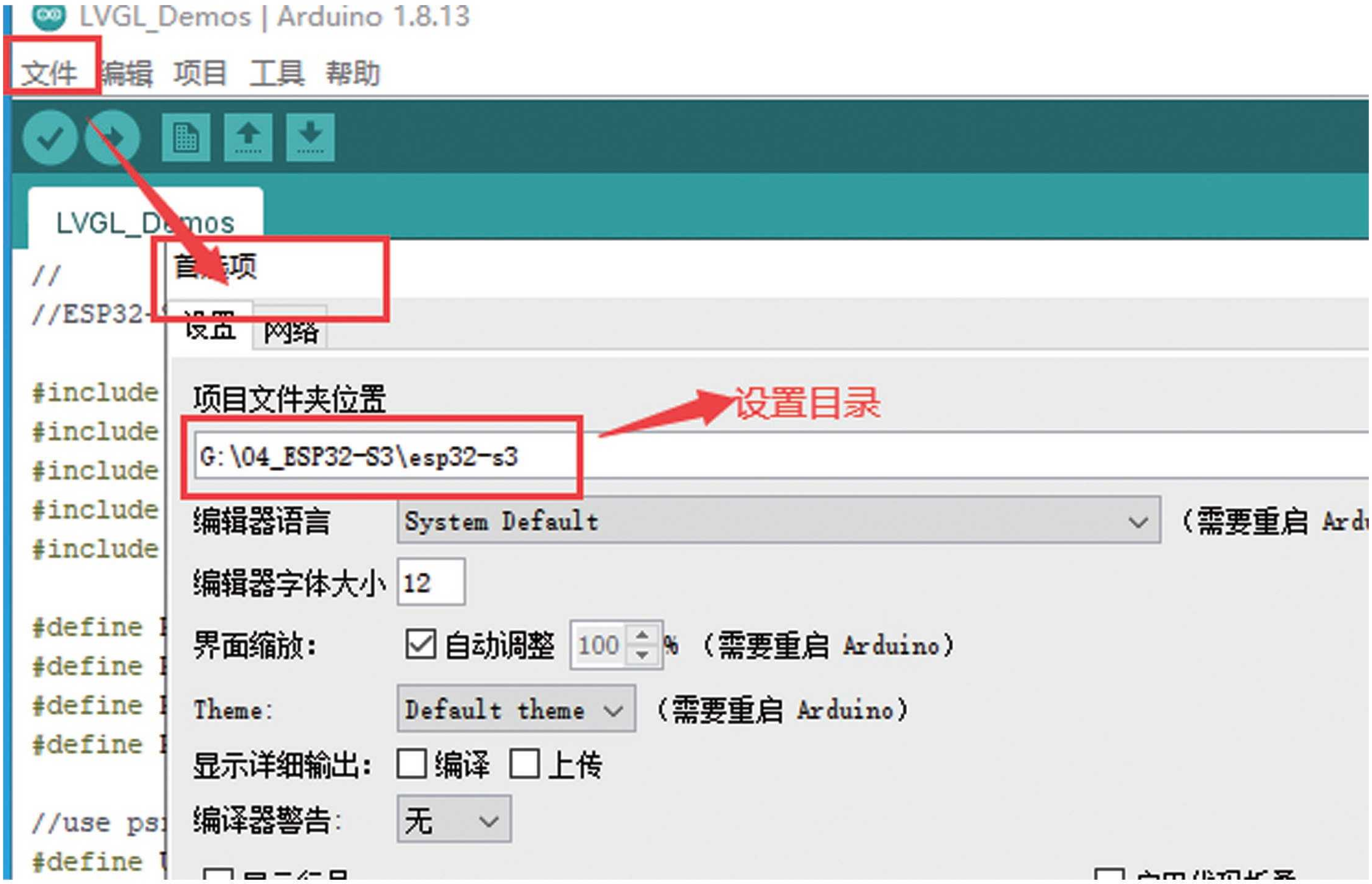


Future video tutorials will be based on the explanation of the expansion board.

1、 Arduino Video Tutorial

a) Copy the dependency library of the routine to the folder

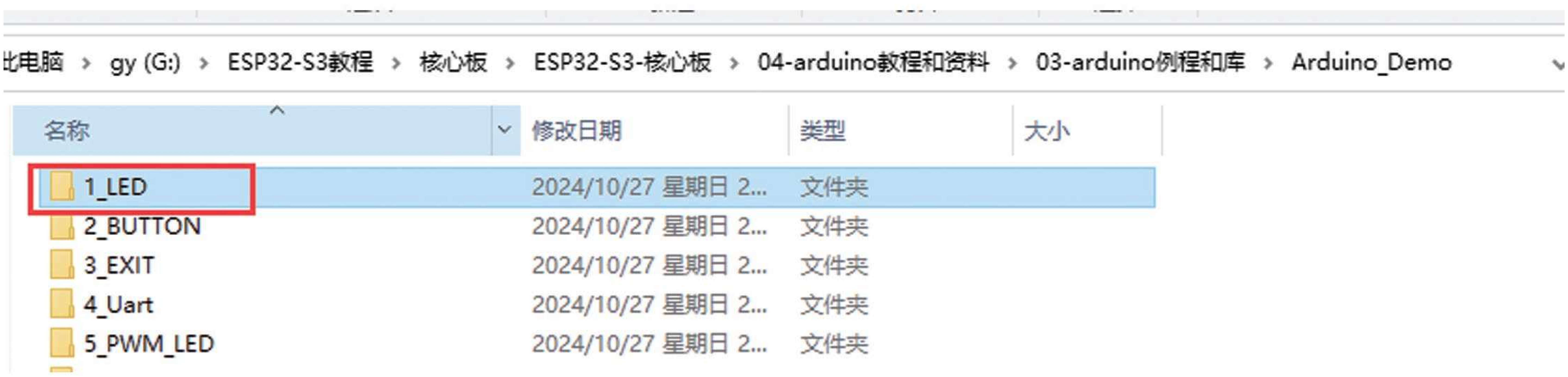


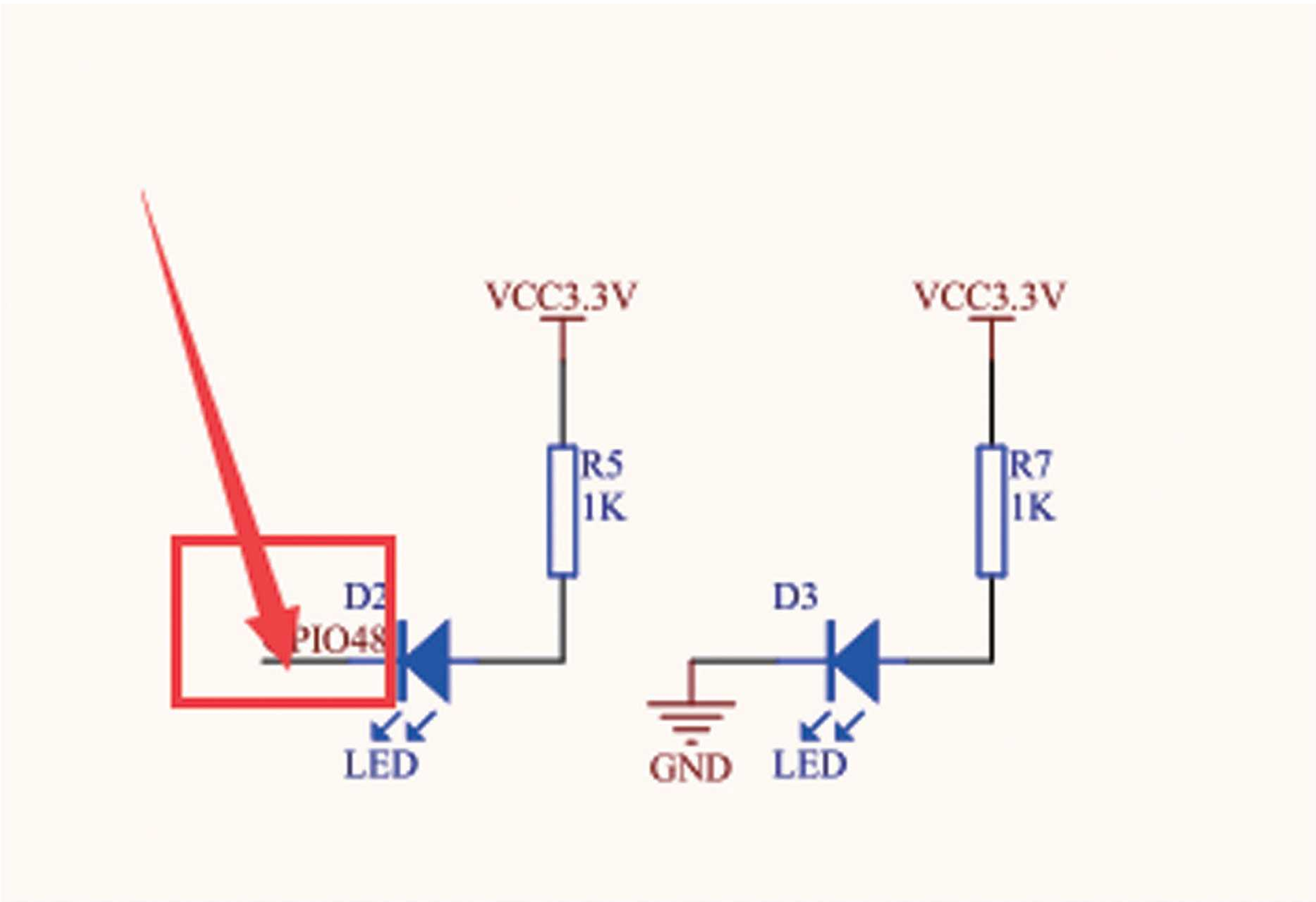


b) Routine Explanation (VSCode+Arduino)

If there is no expansion board, everyone can wire it themselves and connect it to external devices.

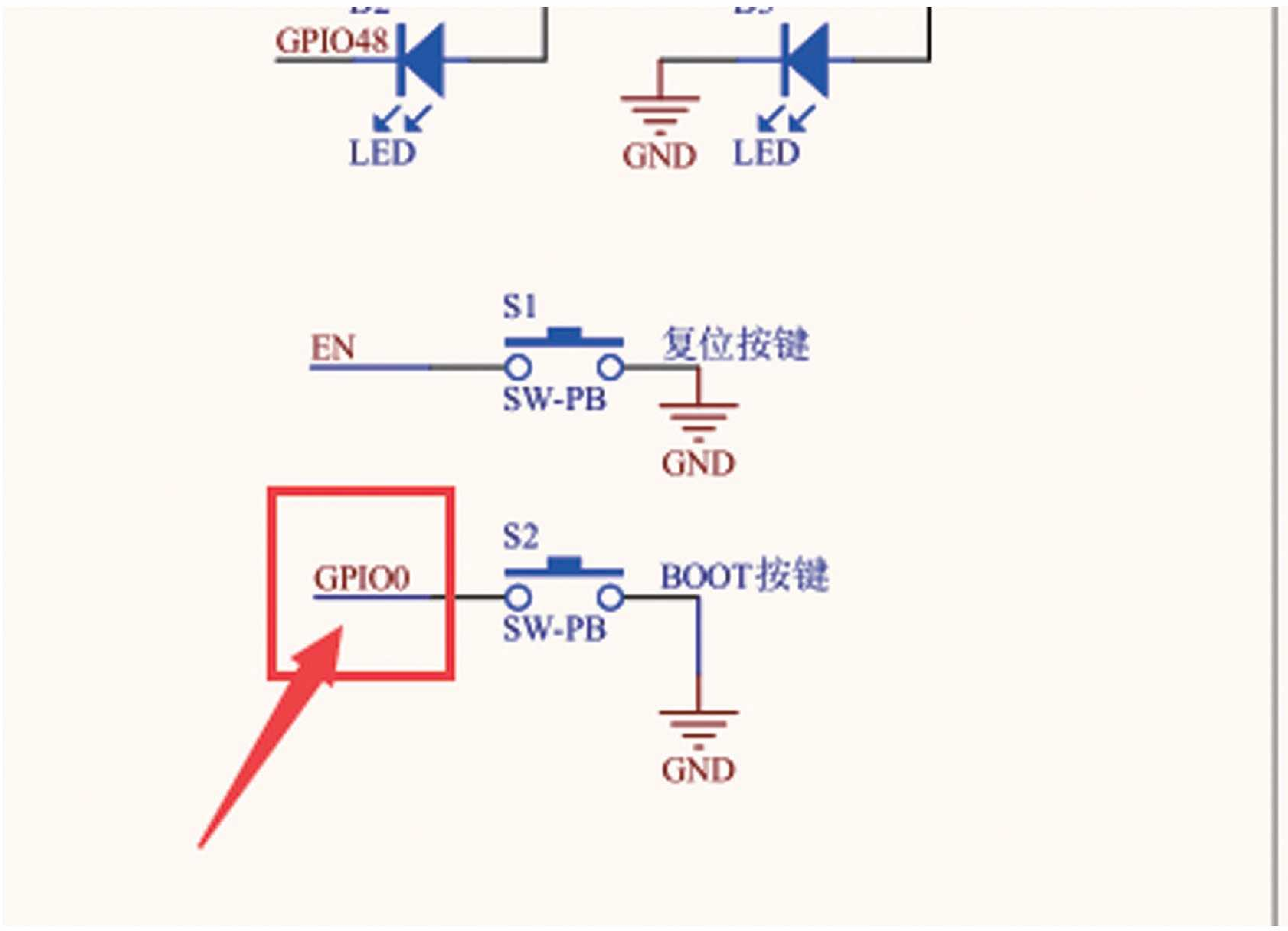
*LED control





*Button routine

名称	修改日期	类型	大小
1_LED	2024/10/27 星期日 2...	文件夹	
2_BUTTON	2024/10/27 星期日 2...	文件夹	
3_EXIT	2024/10/27 星期日 2...	文件夹	
4_Uart	2024/10/27 星期日 2...	文件夹	



Can't interrupt control problem troubleshooting?

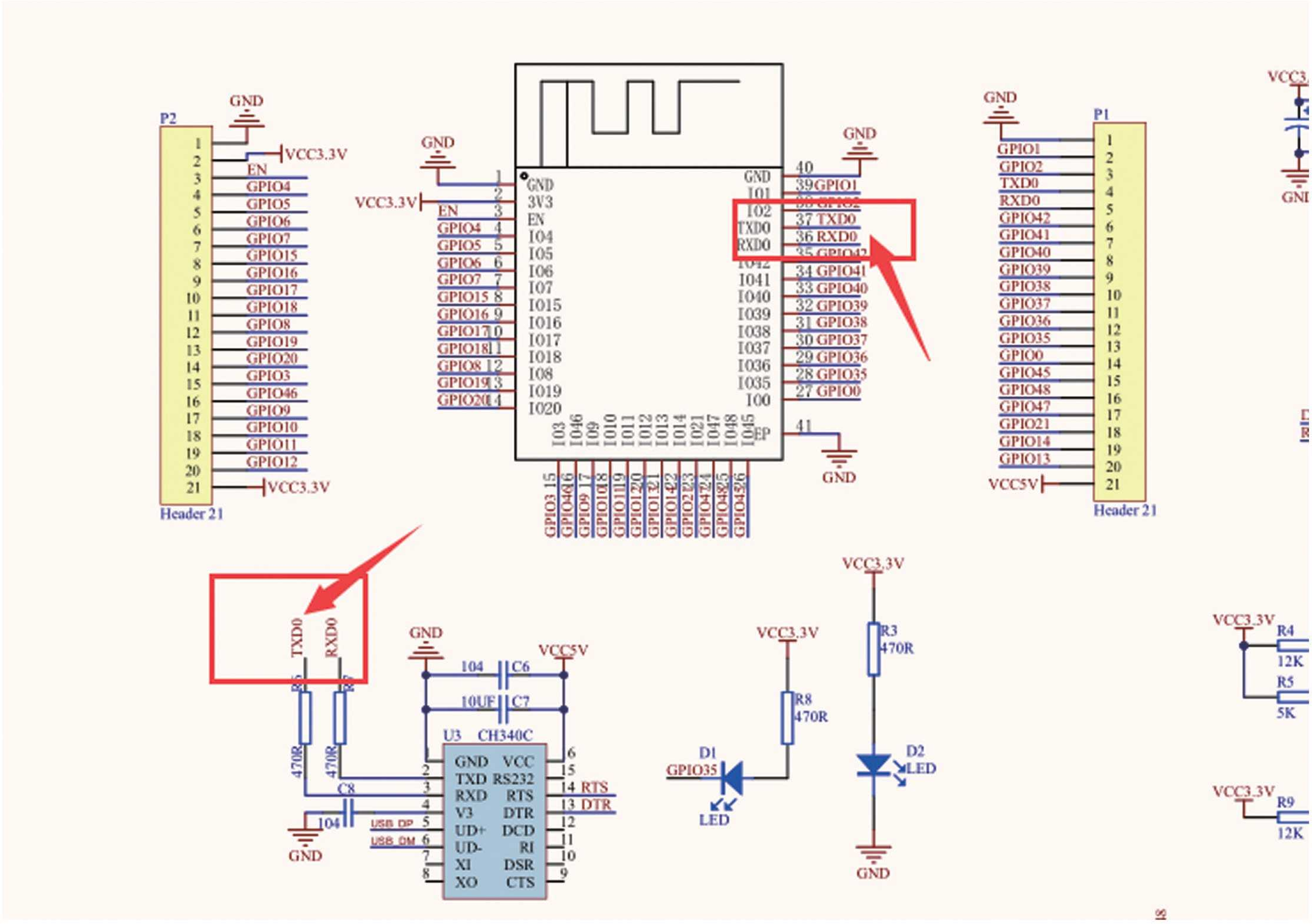
Arduino routines use wiki link address: https://docs.espressif.com/projects/arduino-esp32/en/latest/libraries.html

1) The interrupt function requires the addition of the ARDUINO-ISR-ATTR macro

2) Debugging using serial port

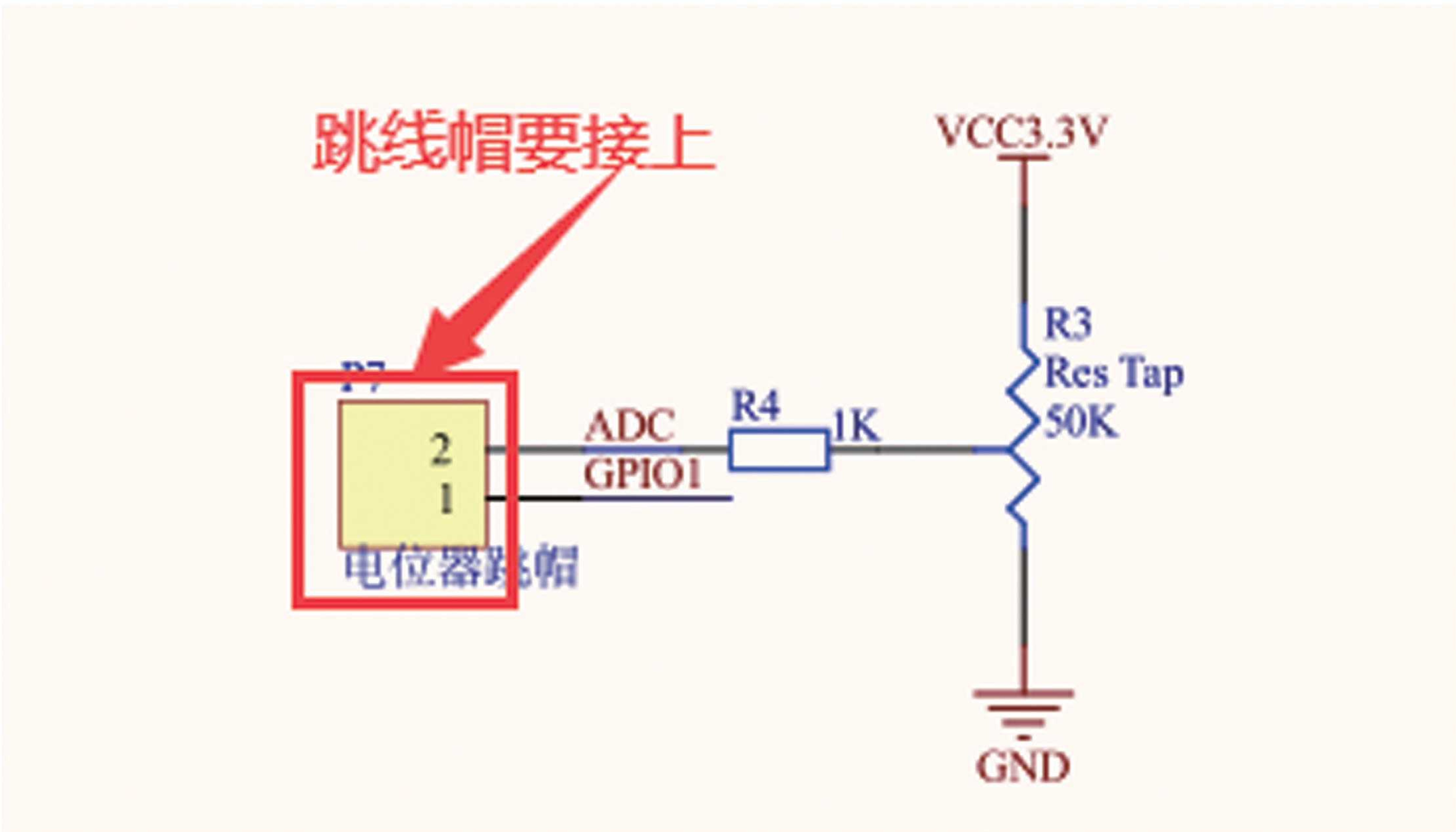
3) Due to incorrect flash size settings, the program ran away during runtime

*Uart routine



The serial port receiver has a buffer, and each read is actually reading data from the buffer, not directly from the serial port

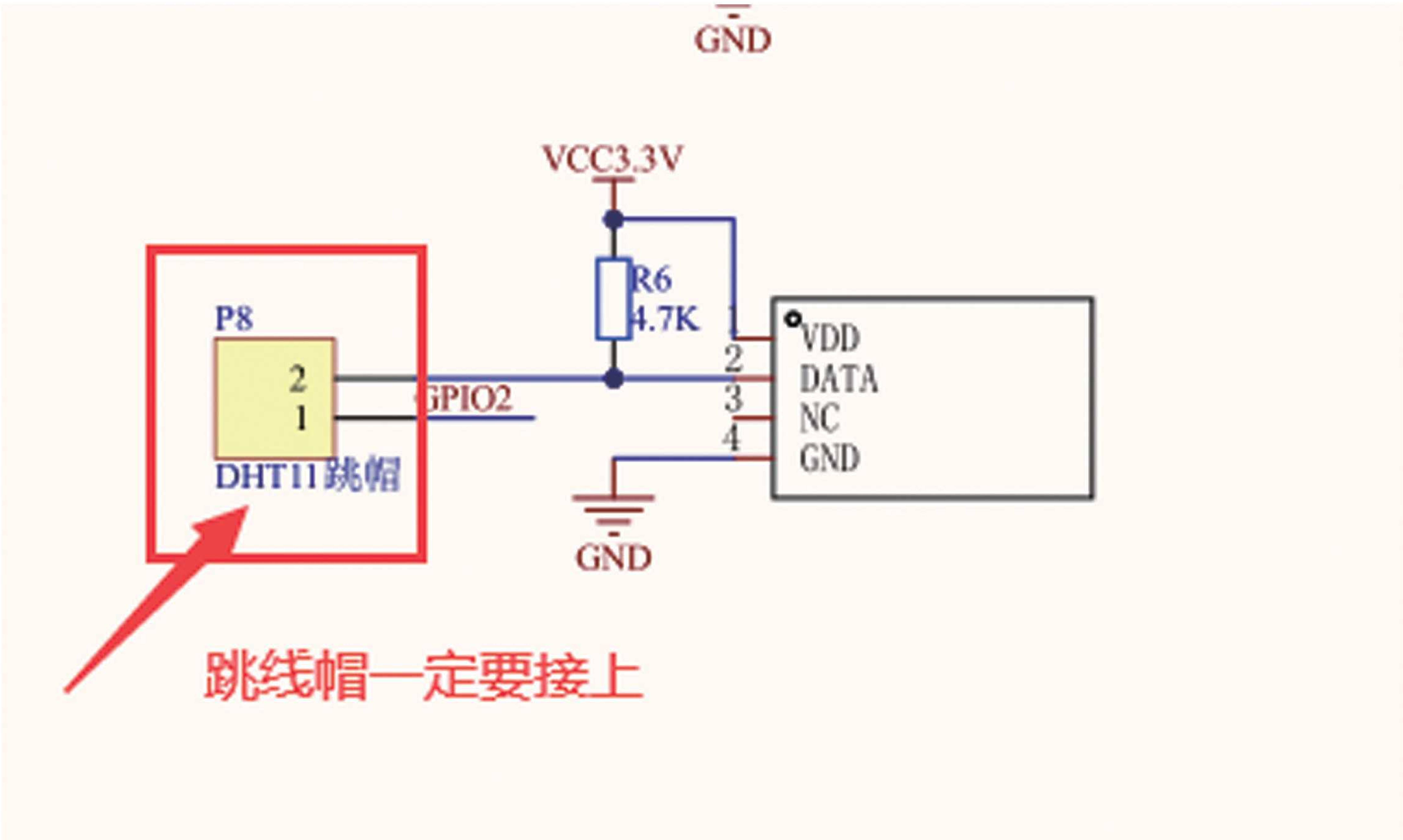
*ADC routine



2^12=4096

GPO1 cannot be reused with other functional pins

*DHT11 Temperature and Humidity Sensor Routine



DHT11 output data format:

8-bit humidity integer data+8-bit humidity decimal data+8-bit temperature integer data+8-bit temperature decimal part+8-bit checksum

byte4	byte3	byte2	byte1	byte0
00101101	00000000	00011100	00000000	01001001
整数	小数	整数	小数	校验和
湿度		温度		校验和

The values of humidity and temperature can be obtained from the above data, and the calculation method is:

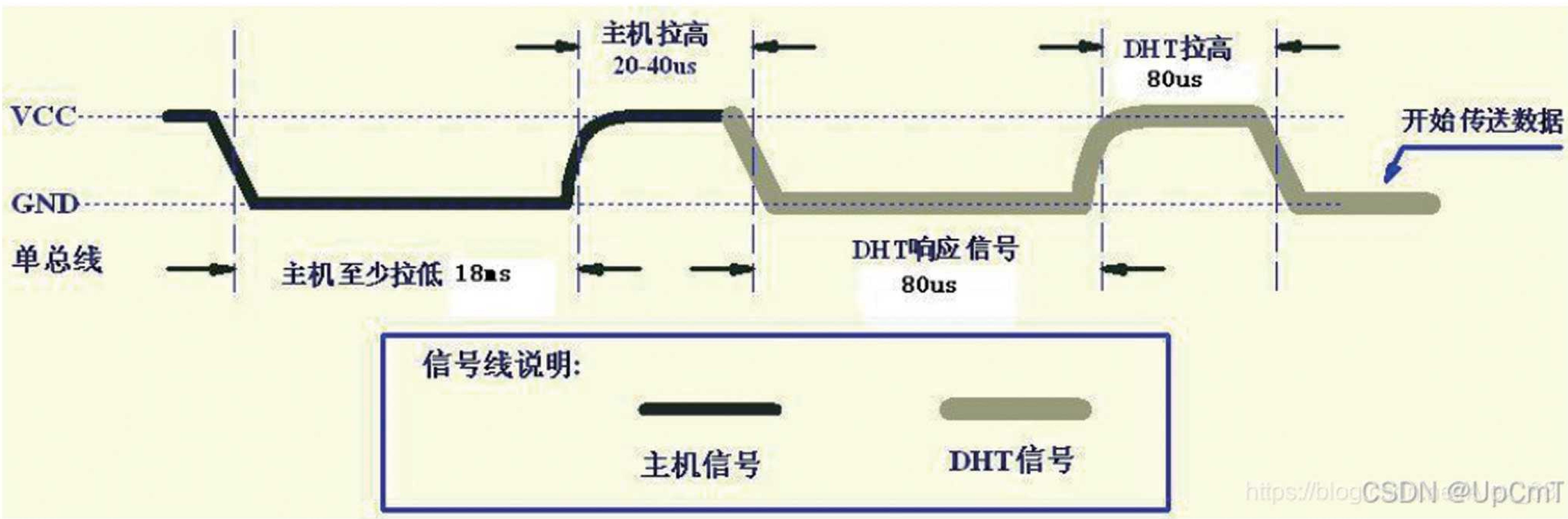
Humidity=byte4 byte3 = 45.0(%RH)

Temperature=byte2 byte1 = 28.0(℃)

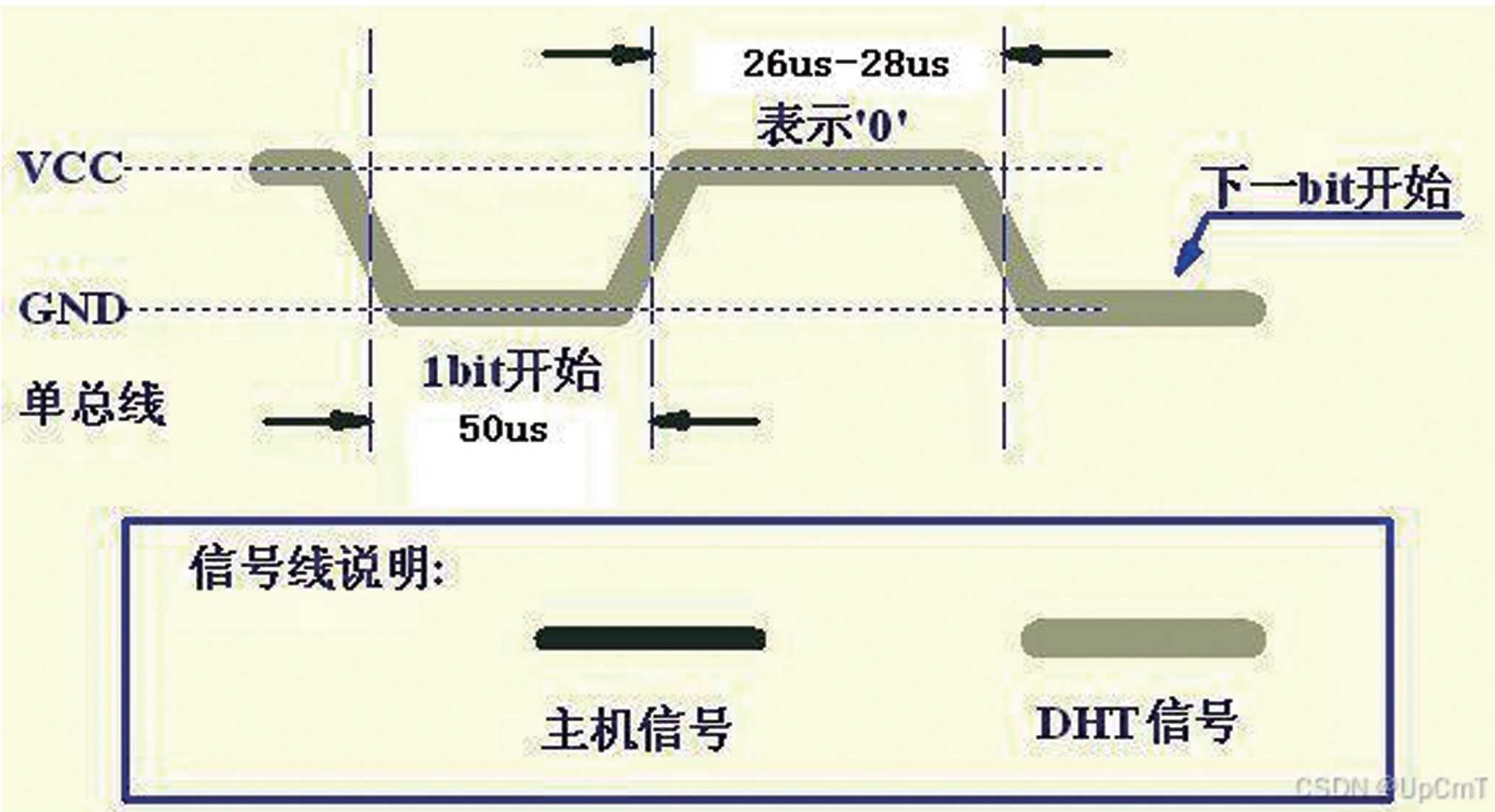
Verification (byte5)=? Byte4+byte3+byte2+byte1=73 (=humidity+temperature) (if they are equal, the verification is correct)

Sequence introduction: The host (ESP32) sends a data transmission command to DHT11 (pull low and then pull high) ->DHT11 responds to the received command (pull low and then pull high pin) ->DHT11 transmits data to the host (ESP32) (pull low and then pull high)

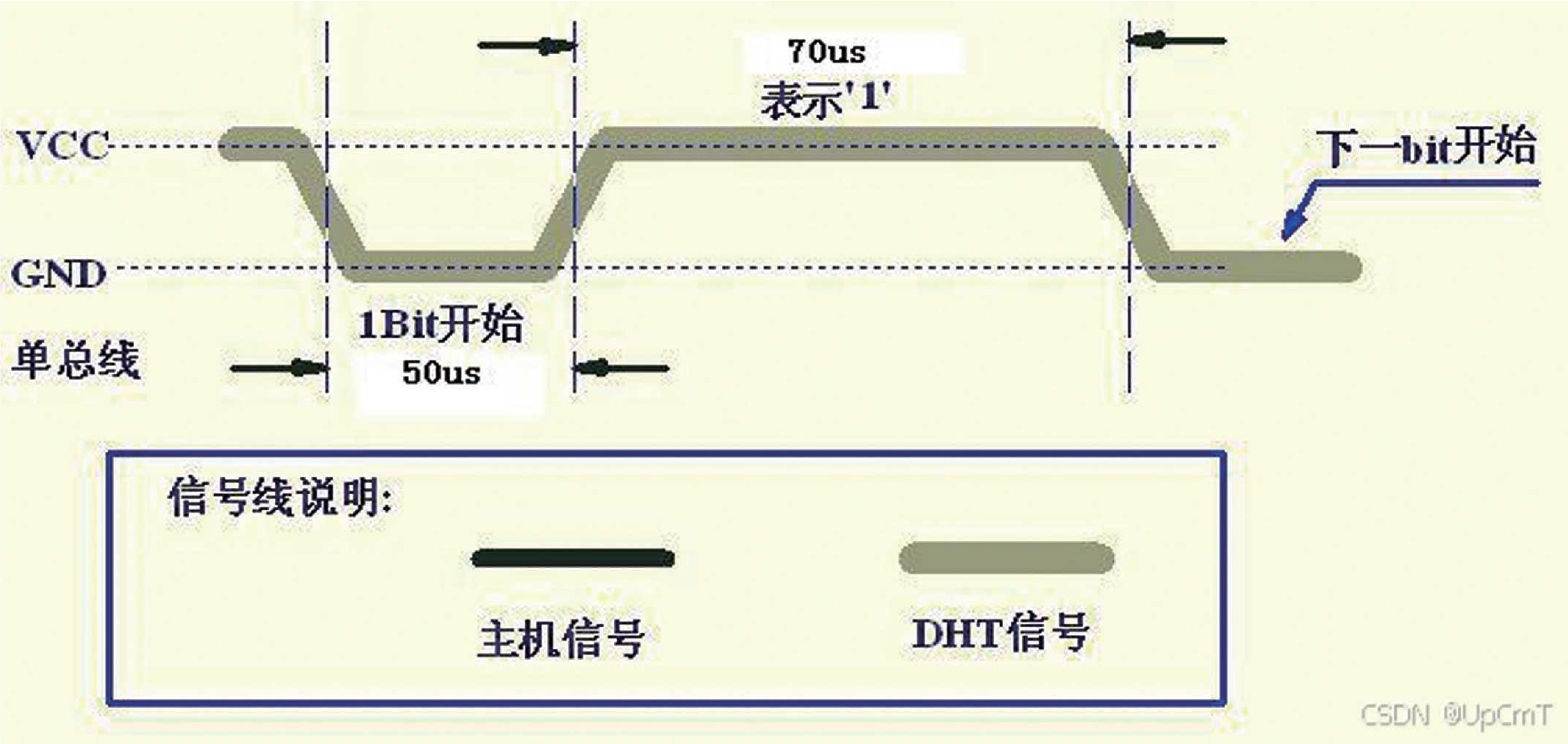
Command sending and response:



Data 0:

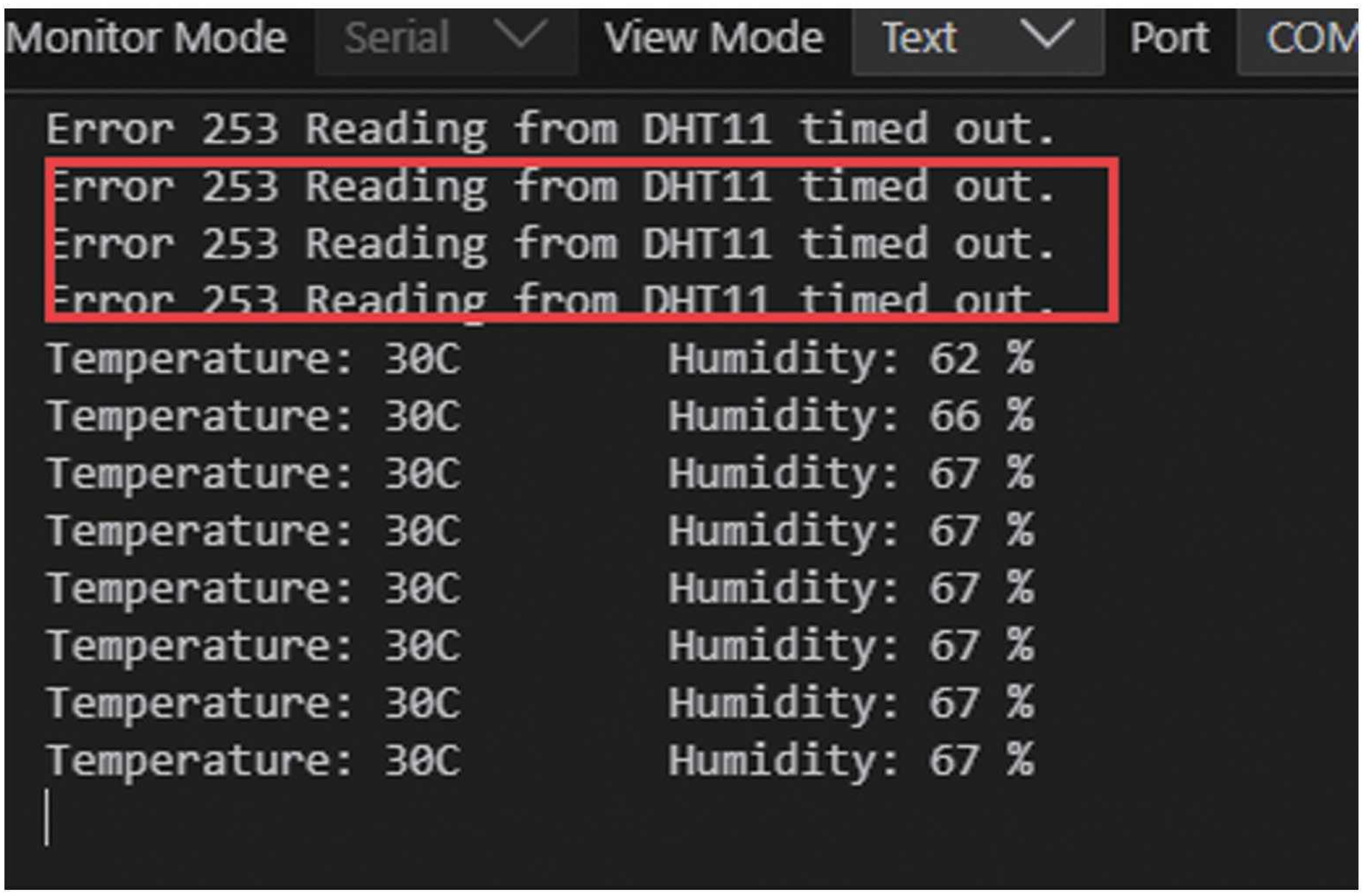


Data 1:



Explanation: The specific value in the above figure is actually an intermediate value taken according to the required time period. You can refer to the manual for the specific range.

Summary of timing: Whether the host sends to DHT11 or DHT11 sends to the host, it is a process of pulling down and then pulling up the pin to complete an action, including commands, responses, and sending 1-bit data. The difference is the time for pulling down and pulling up



*OLED routine

The pins of GPO41 and GPO42 cannot be occupied

Oled Screen Introduction:

1) Screen initialization

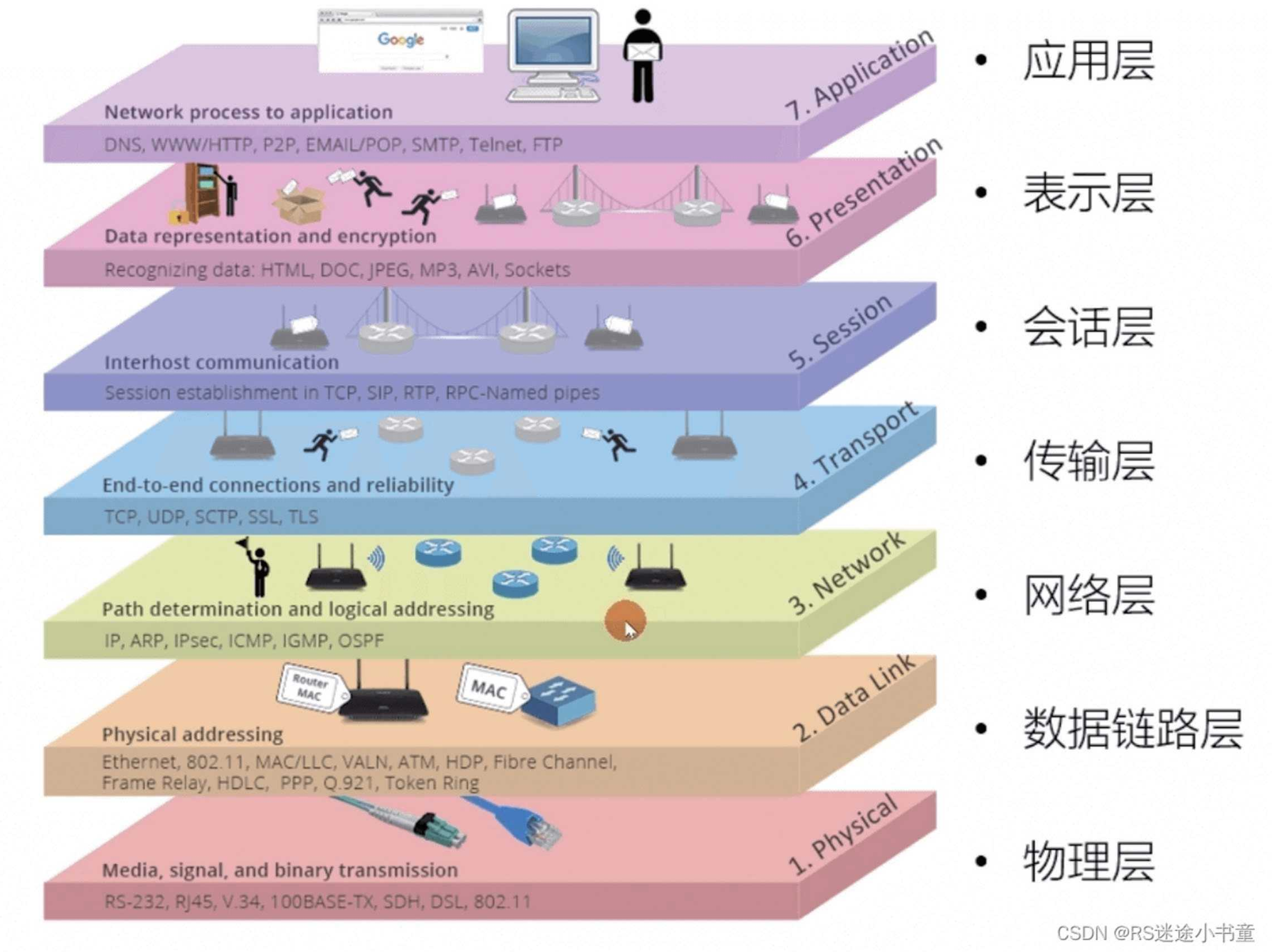
2) Position the cursor on the video memory

3) Write data to the cursor position, and after writing the data, the cursor will automatically move backward by 1 unit

Here is the layout diagram of the video memory: 64X128 bits

64			0 列	1 列	2 列	3 列	....	124 列	125 列	126 列	127 列
Page0	bit0	0行	0								
	bit1	1行	0								
	bit2	2行	0								
	bit3	3行	1								
	bit4	4行	0								
	bit5	5行	0								
	bit6	6行	0								
	bit7	7行	0								
Page1		8到15行									
Page2		16到23行									
Page3		....									
Page4		....									
Page5		....									
Page6		....									
Page7		56行到63行									

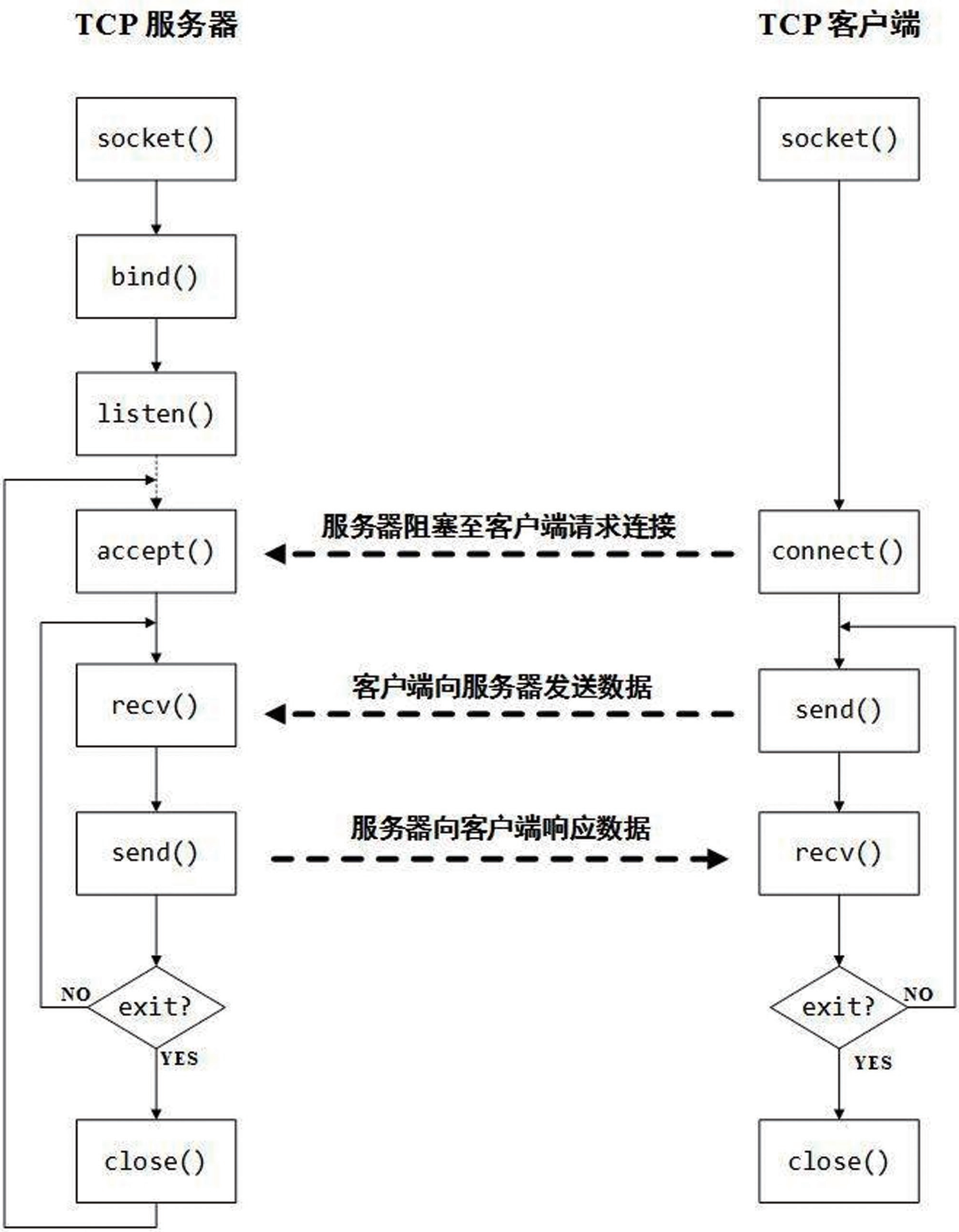
3) The data written to the OLED screen is divided into two types,
Command type: such as controlling cursor position, 0x00+command value
Data type: such as write cached value, 0x01+data value
4) Display principle: Allocate a buffer of the same size as the screen [size: 64x128/8 bytes], modify the contents of the buffer, and directly send all the contents of the buffer to the video memory, refreshing all columns of the video memory
Wifi routine
Seven layer network model:



Memory method: "Internet of Things, Numbers, Communication, Meeting, and Correspondence". Remember these 7 words, and you will remember these 7 layers
conform to

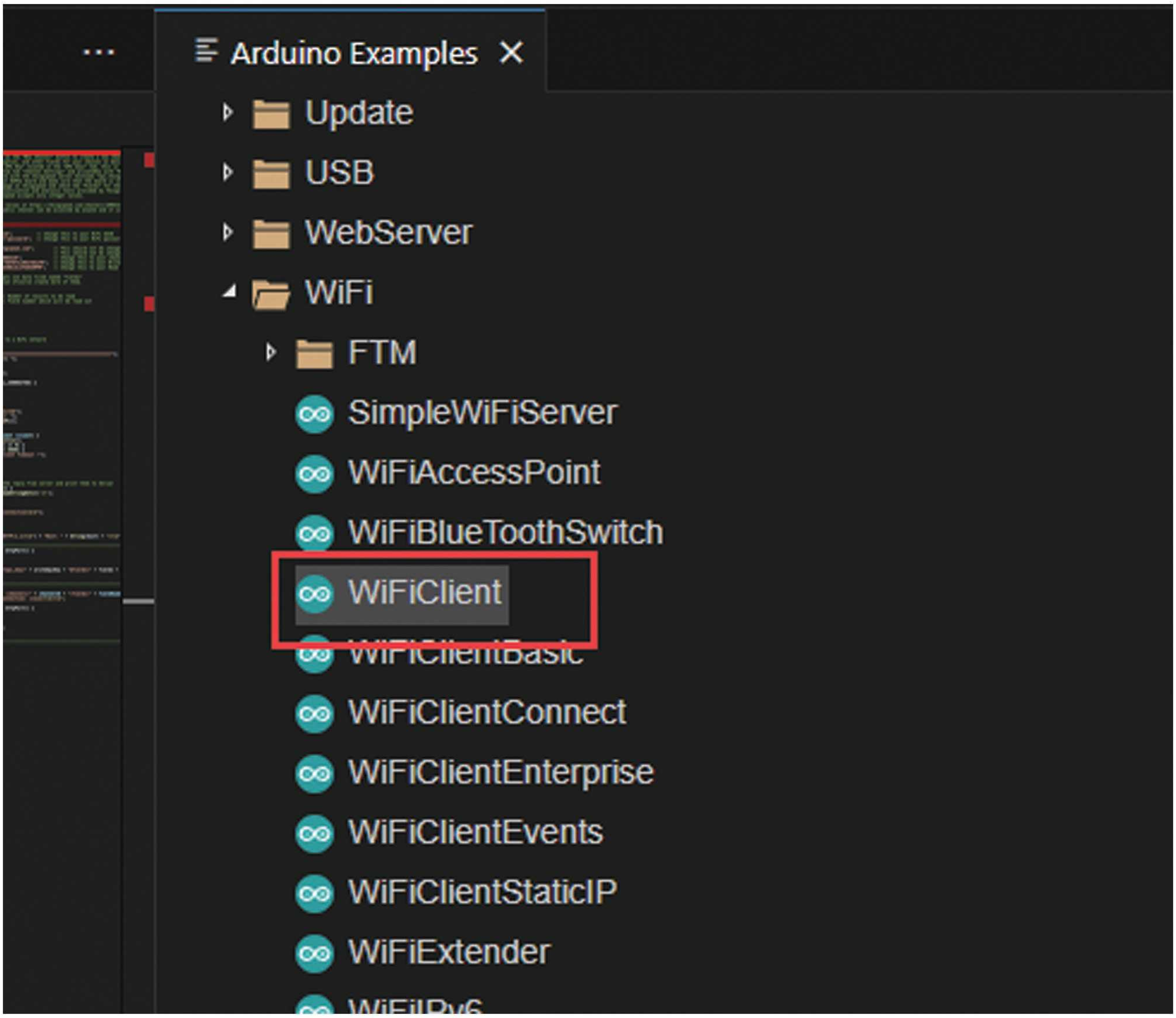
The data is ultimately transmitted to the hardware network through the data link layer and physical layer, and the data at both the data link layer and physical layer is encapsulated layer by layer.

1) TCP communication process



TCP 服务器
TCP 客户端
socket()
bind()
listen()
accept()
recv()
send()
exit?
NO
YES
close()
socket()
connect()
send()
recv()
exit?
NO
YES
close()
服务器阻塞至客户端请求连接
客户端向服务器发送数据
服务器向客户端响应数据

2） tcp client



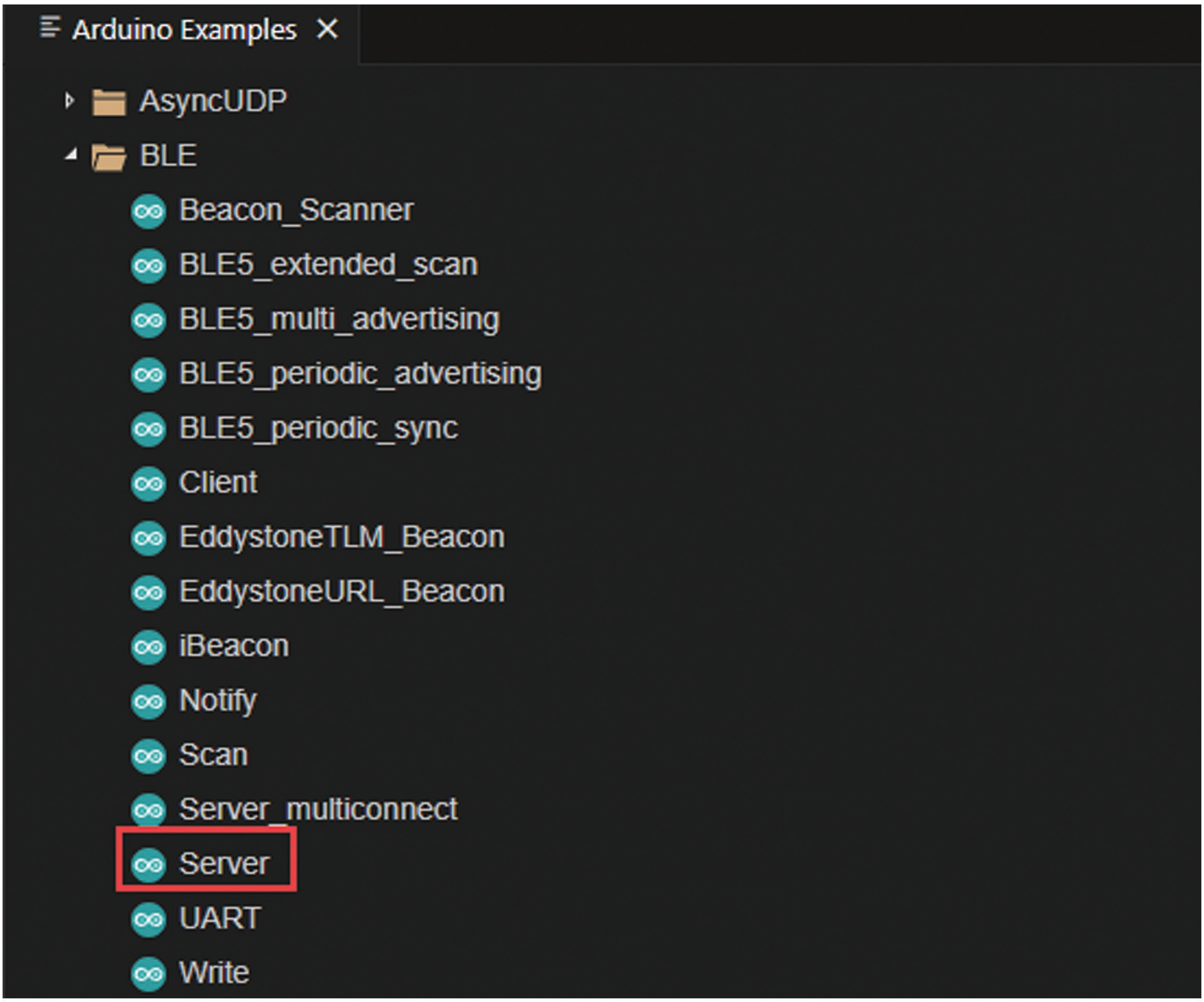
HTTP and MQTT are protocols built on top of TCP.

*BLE routine

Our previous tutorial, if you are interested, you can watch it. If you have any questions, you can communicate in the group.

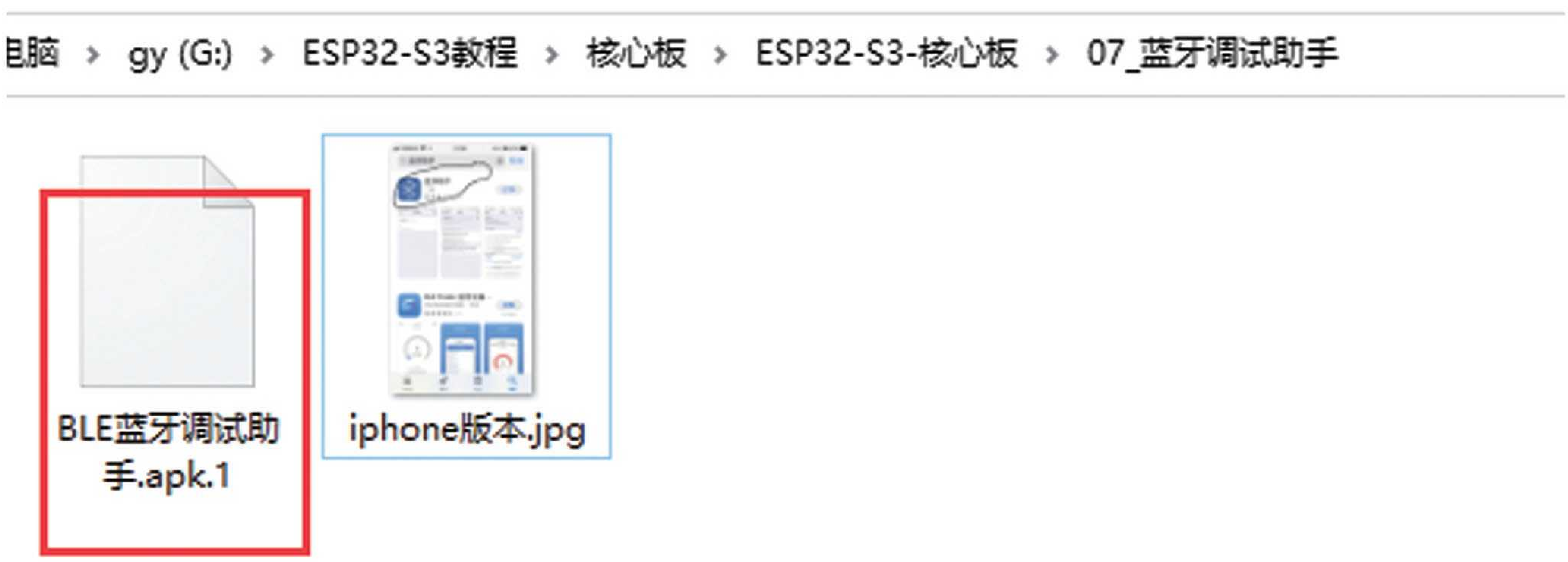
Reuse of previous tutorials,

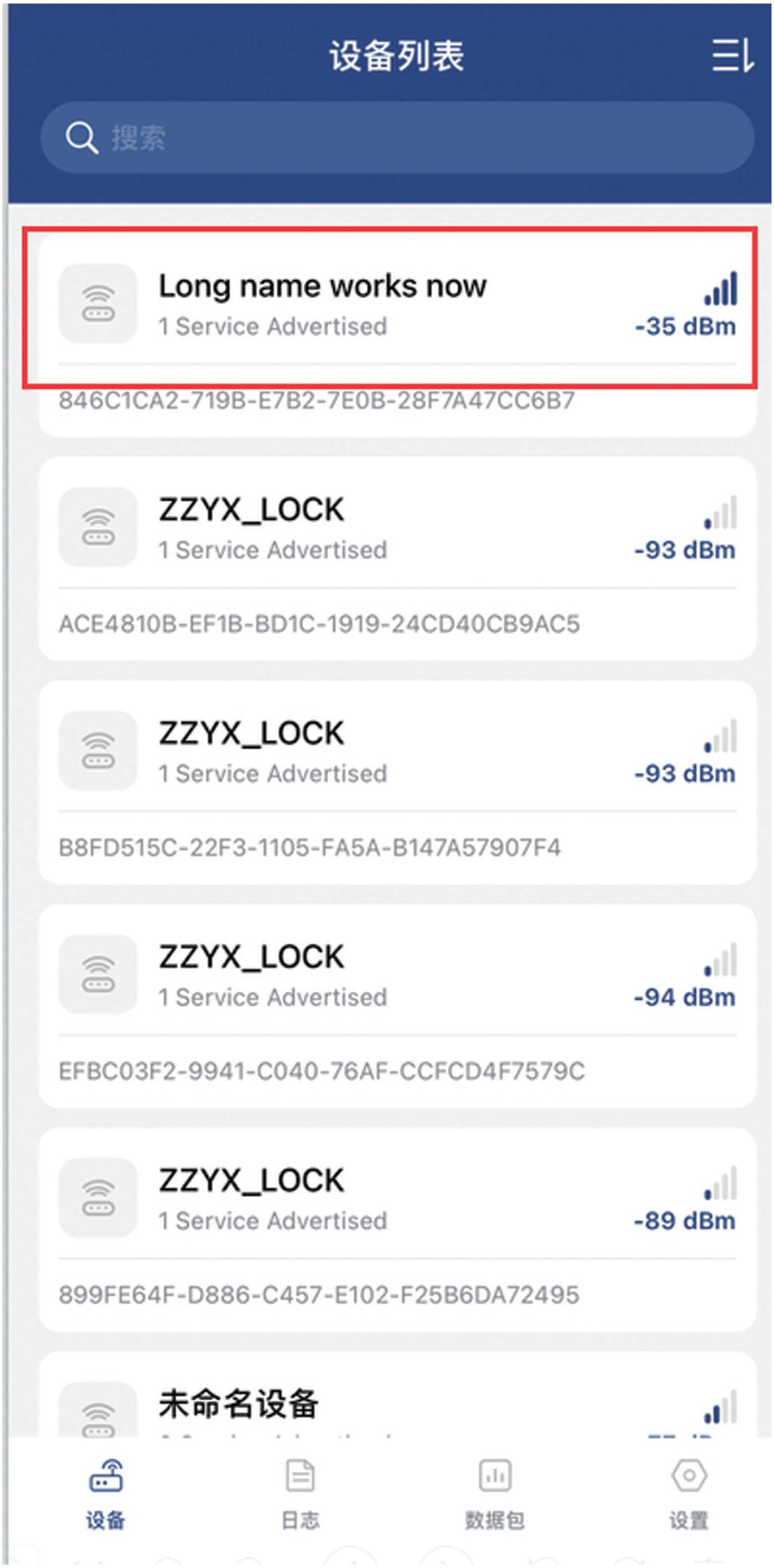

01-arduino依赖库copy.mp4	2024/10/27 星期日 2...	MP4 文件	11,304 KB
02-led例程讲解.mp4	2024/10/27 星期日 2...	MP4 文件	21,363 KB
03-Button例程.mp4	2024/10/27 星期日 2...	MP4 文件	38,523 KB
04-排查按键中断不生效问题.mp4	2024/11/1 星期五 2:07	MP4 文件	39,344 KB
05-串口例程演示.mp4	2024/11/3 星期日 22:...	MP4 文件	75,627 KB
06-ADC例程演示.mp4	2024/11/3 星期日 22:...	MP4 文件	33,243 KB
07-DHT11数据读取介绍.mp4	2024/11/3 星期日 23:...	MP4 文件	28,285 KB
08-DHT11例程分析.mp4	2024/11/3 星期日 23:...	MP4 文件	37,707 KB
09-oled例程分析.mp4	2024/11/4 星期一 0:27	MP4 文件	68,381 KB
10-网络通讯协议介绍.mp4	2024/11/4 星期一 1:06	MP4 文件	44,753 KB
11-wificlient分析.mp4	2024/11/4 星期一 1:43	MP4 文件	59,976 KB
12-蓝牙server相关的使用之一.wmv	2024/11/4 星期一 1:50	WMV 文件	206,218 KB
13-通知怎么被订阅以及server实验.wmv	2024/11/4 星期一 1:50	WMV 文件	161,741 KB
14-service的特征值的读写.wmv	2024/11/4 星期一 1:48	WMV 文件	52,790 KB
15-蓝牙广播和连接理论知识.wmv	2024/11/4 星期一 1:50	WMV 文件	205,992 KB
16-client编程思路.wmv	2024/11/4 星期一 1:48	WMV 文件	23,024 KB
17-client编程实现代码分析.wmv	2024/11/4 星期一 1:50	WMV 文件	153,651 KB
18-怎么获取server的service和characteristic.wmv	2024/11/4 星期一 1:50	WMV 文件	108,149 KB



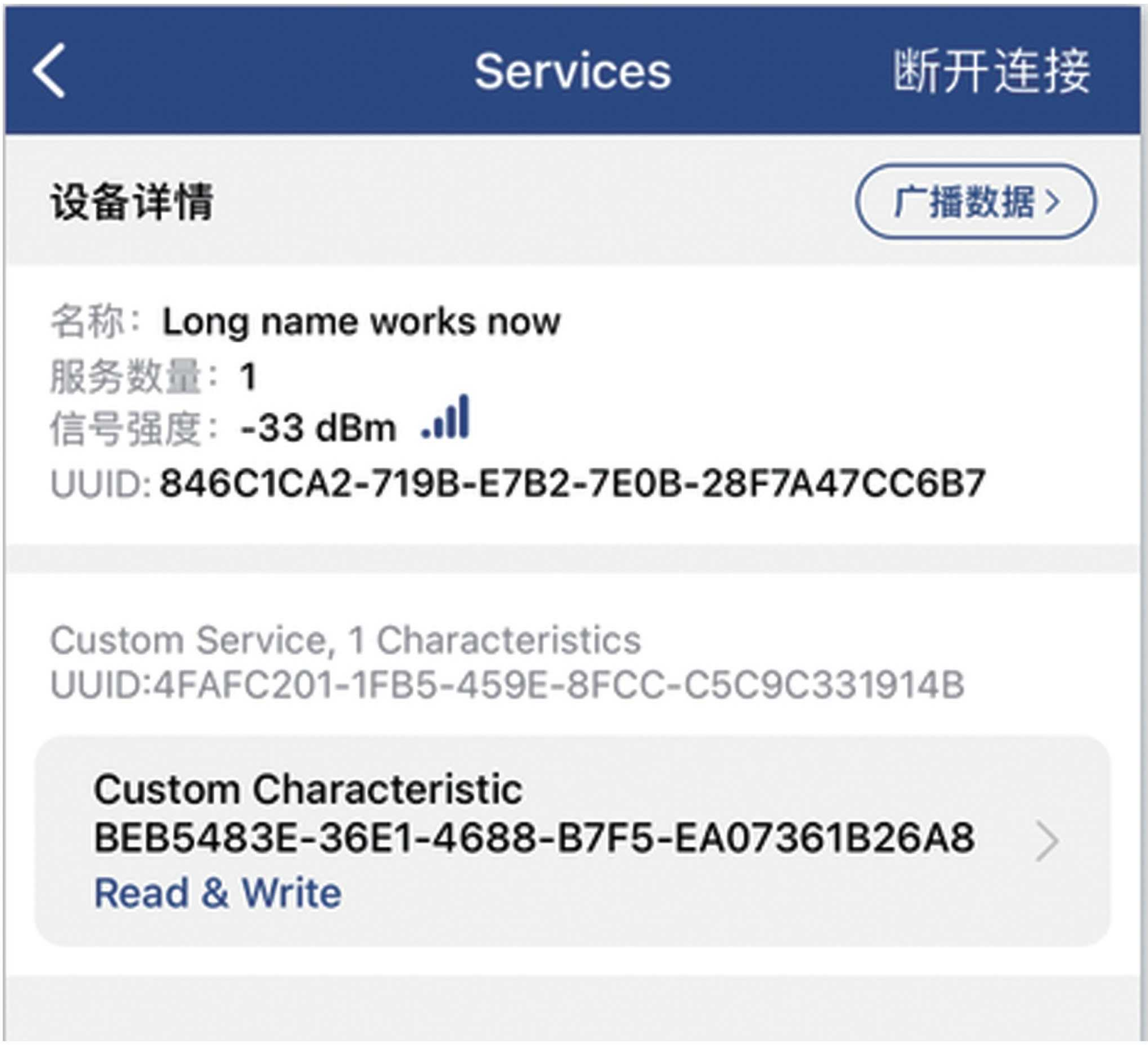
Apple phones can download the following software for debugging:

Android can directly install software:




设备列表
搜索
Long name works now
1 Service Advertised
-35 dBm
846C1CA2-719B-E7B2-7E0B-28F7A47CC6B7
ZZYX_LOCK
1 Service Advertised
-93 dBm
ACE4810B-EF1B-BD1C-1919-24CD40CB9AC5
ZZYX_LOCK
1 Service Advertised
-93 dBm
B8FD515C-22F3-1105-FA5A-B147A57907F4
ZZYX_LOCK
1 Service Advertised
-94 dBm
EFBC03F2-9941-C040-76AF-CCFCD4F7579C
ZZYX_LOCK
1 Service Advertised
-89 dBm
899FE64F-D886-C457-E102-F25B6DA72495
未命名设备
设备
日志
数据包
设置

After clicking on the Bluetooth name, it will automatically connect to Bluetooth:



Click on 'Custom Characteristic' and select 'read' to read the user-defined values,

After writing it, reading it out again is the value written by the user,

*About TFT LCD Video Tutorial Instructions

The Lcd routine will be produced in conjunction with the vlgl tutorial!!

2、 Micropathon Video Tutorial

3、 Idf Video Tutorial